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The MUSIC Critical Benchmark and Nuclear Data

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Abstract — *The Measurement of Uranium Subcritical and Critical (MUSIC) experiment was a series of measurements of critical and subcritical configurations of bare highly enriched uranium. The goal was to compare measurement methods, analysis techniques, and simulation methods across regimes of criticality and to provide high-quality validation of ^{235}U nuclear data. A benchmark evaluation of the two critical configurations of the MUSIC experiments will soon be published in the release of the International Criticality Safety Benchmark Evaluation Project Handbook. The recent execution of the experiment aids in proper quantification of model simplifications and all uncertainties associated with the experiment. Historical benchmark evaluations are heavily relied on for uranium nuclear data validation despite the fact that the same level of documentation and comparable uncertainty analysis may not be present. The MUSIC evaluation is less likely to include "unknown unknowns" that could impede accurately modeling the system. Presented are both highly detailed and very simplified models, which represent the experimental configurations accurately, aiding the users of the benchmark for nuclear data or transport code validation. The sensitivities of k_{eff} to nuclear data and nuclear data–related uncertainties are very similar between this experiment and previous bare uranium sphere experiments. In addition, the nuclear data uncertainties to any nuclides other than ^{235}U are small. For all these reasons, the recently evaluated MUSIC benchmark critical configurations could prove very useful for ^{235}U nuclear data validation. Currently, major libraries have good agreement with the experimental results, within 200 pcm for all nuclear data libraries, and within one standard deviation of the experimental result for most. Suggested nuclear data adjustments based on MUSIC and Lady Godiva are also presented, with posterior improvements to both the agreement in k_{eff} and the uncertainty associated with the nuclear data.*

Keywords — *Critical experiments, uranium, nuclear data, benchmark.*

Note — *Some figures may be in color only in the electronic version.*

I. INTRODUCTION

One of the primary methods for validation of both computational methods and nuclear data is comparison of calculated results using models of integral experiments to experimental results. The International Criticality Safety Benchmark Evaluation Project (ICSBEP) Handbook is a repository of many such evaluated experiments. Recent releases of nuclear data libraries have used

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improvements in the agreement between the simulation and the experiments to justify changes to the cross-section evaluations.[1] Experiments are selected that are sensitive to a given nuclear interaction [e.g., the neutron-induced fission cross section for ^{235}U expressed by $^{235}\mathrm{U}(n,f)$], and the agreement between the simulated result and the experimental result is evaluated.

Experiments are approved as benchmarks after review by the ICSBEP Technical Review Group (TRG). The TRG determines if all known sources of uncertainty and potential biases have been adequately evaluated. This implies that any remaining discrepancy could be attributed to the nuclear data or errors in computation. Over the past few decades, the standards have increased of what constitutes an acceptable evaluation and, in particular, an acceptable uncertainty analysis. Many of the early experiments approved as benchmarks were not evaluated at the same level of detail, especially in the uncertainty analysis, as more recent benchmarks.

It is difficult, if not impossible, to update the evaluation of some experiments in part because these experiments may have been performed decades before the evaluation was written and the components may no longer exist. In the time gap between experimental execution and evaluation, documentation for parameters of interest may not exist or have been lost, which increases uncertainty.[2] Thus, it is likely that there are "unknown unknowns," potential sources of bias or uncertainty that are not addressed in older benchmarks.

Modern benchmark experiments are less likely to have this issue, as the uncertainty analysis tends to be much more thorough, they are documented with the intention of being evaluated, and the people who performed the experiment are available to answer questions or obtain data necessary for a more detailed analysis.

The Measurement of Uranium Subcritical and Critical (MUSIC) experimental campaign was a series of subcritical and critical experiments executed at the National Criticality Experiments Research Center located at the Nevada National Security Site between December 2020 and March 2021. The series was a collaboration between Los Alamos National Laboratory (LANL), the French Institut de Radioprotection et Sûreté Nucléaire, and the University of Michigan. MUSIC consisted of ten configurations composed of highly enriched uranium (HEU) hemishells arranged in spherical geometries at neutron multiplications between deeply subcritical and delayed supercritical. The primary goal was to test a variety of different detection systems, analysis methods, and simulation techniques across a wide range of multiplication regimes.

Another important objective was to provide a modern set of experimental measurements with better quantified uncertainties and biases to more accurately validate ^{235}U neutron cross-section data. Previous evaluations of ^{235}U suffer from potentially incomplete documentation.[3,4] Lady Godiva was performed in the 1950s, and the entire uncertainty analysis is only two pages of text and uses methods to give the uncertainty in the critical mass as opposed to the modern methods of propagating uncertainties in physical parameters to k_{eff} uncertainties.

The ORSphere benchmark is to a more modern standard, having been written in 2014; however, the experiment itself was performed in the 1970s. Because of this, an experiment both performed in recent memory and documenting the latest in benchmark standards would help ascertain whether the benchmark uncertainties posed by the earlier bare sphere benchmarks are reliable.

The series measured ten different configurations using HEU hemishells, known as the Rocky Flats (RF) shells. Eight of them were subcritical,[5,6] and two of them were supercritical.[7] This paper will focus on the two supercritical configurations (Case 1 and Case 2). The subcritical configurations will be featured in future publications.

This paper is organized as follows. Section II describes the two supercritical configurations and the evaluated k_{eff} value, which, in this context, is the effective neutron multiplication factor inferred from the measured reactor periods. Section III discusses the evaluation of the uncertainties for the two supercritical configurations and the biases associated with each experiment. The experimental and benchmark k_{eff} values are also presented in Sec. IV, where the benchmark k_{eff} is the value any simulations are expected to match. This is the experimental k_{eff} modified by the effect of any simplifications to the model. Section V discusses the sensitivity of ^{235}U nuclear data and their uncertainty. Finally, Sec. VII provides a summary and conclusions of the benchmark evaluation and its uncertainties as well as an assessment of the sensitivity of the ^{235}U neutron cross-section data to this and other similar benchmarks.

II. DESCRIPTION OF EXPERIMENTAL CONFIGURATIONS AND THEIR k_{eff} VALUES

Both configurations used the RF hemishells that are nesting spherical hemishells of HEU. For safe assembly, the experiment is separated into an upper stack and a lower stack. The lower stack sits on a movable platen while the upper stack sits on a thin stainless steel membrane plate. The Planet vertical-lift assembly machine is used to remotely move the lower stack to meet the upper

stack for final assembly.[8] A picture of a fully assembled subcritical configuration is shown in Fig. 1, while Fig. 2 shows a critical configuration separated into the upper and lower stacks.

The two configurations evaluated in the ICSBEP handbook differ in the measured reactor period when fully assembled. The measured reactor period allows calculation of an excess reactivity, or reactivity above delayed critical, through the Inhour equation. This will be further discussed in Sec. II.D. To experimentally build systems with different excess reactivities, two different subsets of RF hemishells were used. Specifically, varying the total HEU mass and substituting interior RF shells for aluminum mock shells reduced the excess reactivity of the system. The machine can provide fine reactivity control through separation of the two halves, but a method that is more repeatable and easier to evaluate is to use a physical barrier to set a known separation distance. Aluminum spacer rings (see Fig. 2) were placed on the bottom stack to supply a fixed gap between the upper and lower stacks when they are in full contact.

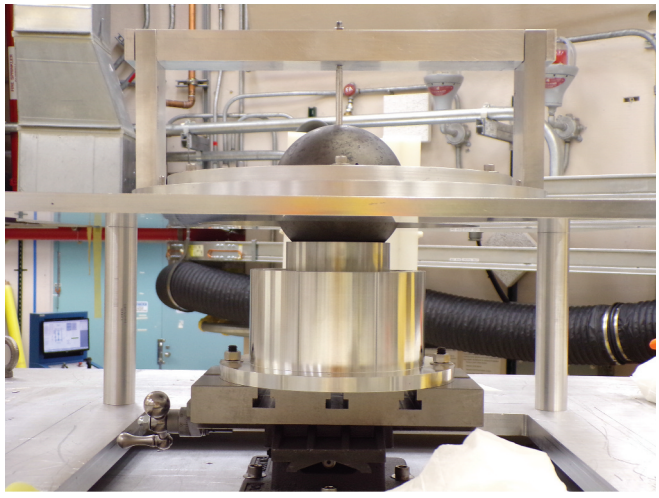


Fig. 1. A view of fully assembled subcritical configuration.

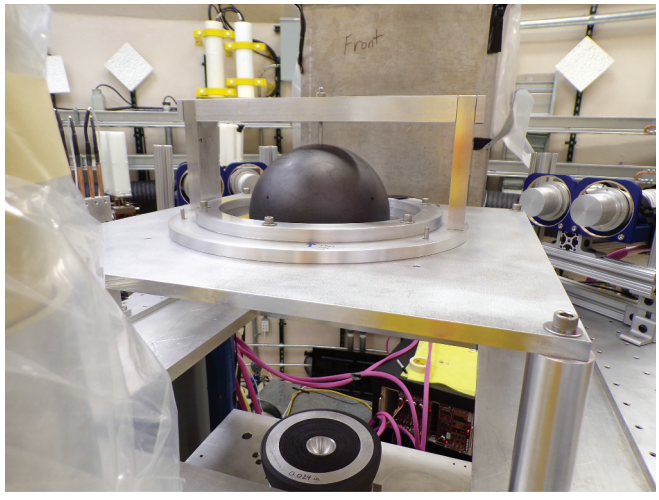


Fig. 2. A view of a critical configuration while fully separated.

Table I lists the part numbers of the RF hemishells, the mock aluminum hemishells, and the spacer ring thicknesses used in both configurations. Even-numbered RF hemishells were positioned on the bottom stack while the odd-numbered RF hemishells were on the top stack for both configurations. For the second configuration, a mock aluminum hemishell is used instead of RF hemishell number 4 (the innermost RF hemishell) to reduce the HEU mass and therefore the excess reactivity. More details about the components, or about other aspects of the assembly, can be found in Ref. [9].

II.A. Rocky Flats Hemishells

The RF hemishells are nesting spherical hemishells fabricated in 1965 for the Rocky Flats Critical Mass Laboratory[10] and have been used as the fissile material in several previous ICSBEP benchmarks.[11–13] Each hemishell is approximately 0.33 cm thick. Each shell is nearly identical in a mating pair of hemishells, at least within machining tolerance, and is labeled with a distinct serial number starting with 1. This creates "odd" and "even" sets of hemishells, such that the odd sets of hemishells nest within each other and the even sets nest within each other. As mentioned previously, the odd hemishells were placed on the upper stack while the even hemishells were placed on the lower stack located on the movable platen.

Figure 3 shows the RF hemishells on a table. Their dimensions and masses measured in 2020 and 2021 (immediately before and after the campaign) are listed in Tables II and III. The average of the mass between these two

TABLE I

Supercritical Configurations of the MUSIC Experiment Campaign

Case	Rocky Flats Shells	Mock Aluminum Shell Number	Spacer Ring Thickness (cm)
1	3-46	–	0.1542
2	3,5-46	4	0.061807



Fig. 3. Rocky Flats hemishells.

measurements was used to calculate the density used in the MCNP® [a] simulations described in Sec. IV. Table IV shows the isotopic composition of the RF hemishells. One measurement was performed in 1965, and another one was performed in 1971. While the ^{233}U was not recorded, Ref. [10] states that multiple measurements were made and that the content was always "less than the detectable limit." The limit was not stated but could safely be assumed to be negligible based on historical familiarity of other HEU made during that time frame.

Each hemishell has five holes: one polar hole and four smaller pry holes. The polar hole facilitated the alignment of the hemishells through the use of an alignment rod. The four smaller pry holes were designed to aid in disassembly if two hemishells became stuck together. The polar holes are approximately 0.714 cm in diameter while the pry holes are 0.318 cm in diameter. The pry holes were drilled parallel to the equator plane and one-third of the outside radius down from this plane toward the pole. Figures 4 and 5 depict the locations of the holes in a drawing and in a close-up picture, respectively.

[a] MCNP® and Monte Carlo N-Particle® are registered trademarks owned by Triad National Security, LLC, manager and operator of LANL. Any third party use of such registered marks should be properly attributed to Triad National Security, LLC, including the use of the designation as appropriate. For the purposes of visual clarity, the registered trademark symbol is assumed for all references to MCNP within the remainder of this paper.

II.B. Aluminum Components

II.B.1. Source Holder

The centralmost component in the assembly was the aluminum source holder, which positioned a neutron

TABLE II

Measured Dimensions for the RF Hemishells from HMF-039

Shell Number	Inner Radius (cm)	Outer Radius (cm)	Shell Number	Inner Radius (cm)	Outer Radius (cm)
3	2.0126	2.3371	4	2.0126	2.3377
5	2.3475	2.6696	6	2.3473	2.6698
7	2.6800	3.0035	8	2.6791	3.0027
9	3.0127	3.3352	10	3.0123	3.3351
11	3.3437	3.6697	12	3.3440	3.6698
13	3.6798	4.0025	14	3.6801	4.0024
15	4.0168	4.3381	16	4.0162	4.3376
17	4.3457	4.6697	18	4.3463	4.6698
19	4.6786	5.0045	20	4.6783	5.0039
21	5.0173	5.3372	22	5.0128	5.3358
23	5.3464	5.6692	24	5.3458	5.6693
25	5.6794	6.0027	26	5.6790	6.0015
27	6.0113	6.3346	28	6.0121	6.3344
29	6.3451	6.6707	30	6.3441	6.6696
31	6.6784	7.0025	32	6.6792	7.0030
33	7.0060	7.3296	34	7.0098	7.3338
35	7.3417	7.6658	36	7.3428	7.6665
37	7.6824	8.0027	38	7.6711	8.0027
39	8.0128	8.3364	40	8.0075	8.3292
41	8.3462	8.6683	42	8.3443	8.6680
43	8.6782	8.9996	44	8.6764	8.9995
45	9.0095	9.3328	46	9.0104	9.3329

TABLE III

Recorded Masses of the RF Shells

Shell Number	2020 Shell Mass (g)	2021 Shell Mass (g)	Shell Number	2020 Shell Mass (g)	2021 Shell Mass (g)
3	173.9	173.9	4	174.5	174.7
5	231.6	231.4	6	232.0	232.0
7	299.8	299.8	8	299.0	299.0
9	373.8	373.8	10	372.7	372.6
11	462.1	462.2	12	462.5	462.5
13	550.3	550.3	14	550.0	550.1
15	647.9	647.9	16	647.2	647.1
17	760.2	760.3	18	760.7	760.6
19	883.2	883.2	20	884.1	883.9
21	995.4	995.3	22	1003.5	1003.3
23	1137.9	1137.7	24	1140.5	1140.3
25	1274.9	1274.6	26	1273.6	1273.5
27	1431.7	1431.8	28	1427.8	1427.8
29	1596.4	1596.2	30	1596.0	1595.8
31	1766.1	1765.8	32	1763.1	1762.5
33	1934.1	1934.0	34	1935.0	1934.8
35	2124.5	2124.0	36	2121.4	2120.8
37	2340.2	2339.7	38	2334.1	2333.3
39	2514.8	2514.7	40	2495.7	2495.7
41	2710.5	2708.8	42	2727.2	2726.8
43	2932.4	2931.8	44	2936.2	2935.7
45	3175.2	3172.5	46	3164.6	3164.1

TABLE IV

Initial Composition of the RF Shells

Isotope	1965 Mass Fraction (%)	1971 Mass Fraction (%)
^{233}U	Not Recorded	Not Recorded
^{234}U	1.00	1.02
^{235}U	93.19	93.16
^{236}U	0.4	0.47
^{238}U	5.41	5.35

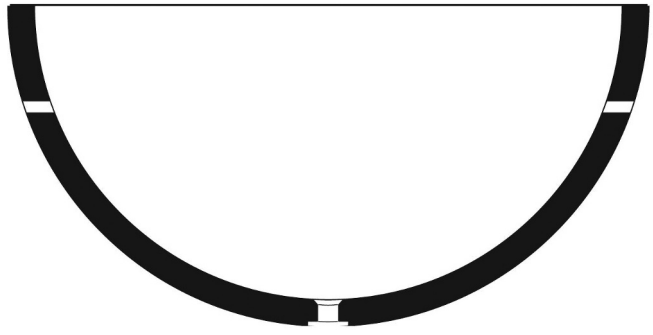


Fig. 4. A representative drawing showing the locations of holes in the RF shells.

source to provide a meaningful indication of the increase in the neutron multiplication during the approach to critical. The holder was composed of two separate components that screwed together. There was a small cavity, or void, inside the two components where a neutron source was placed. The neutron source was removed for the final benchmark k_{eff} measurements in both of the two critical configurations. The source holder was present for all configurations of the experiment.

II.B.2. Mock Aluminum Hemishells

The mock aluminum shells are a set of aluminum spherical hemishells that can be substituted for comparable RF hemishells as a means of adjusting the amount of HEU mass and therefore the reactivity of the system. No mock aluminum hemishells were used for the first critical configuration. For the second configuration, the innermost RF hemishell, number 4, was replaced by a mock aluminum hemishell.

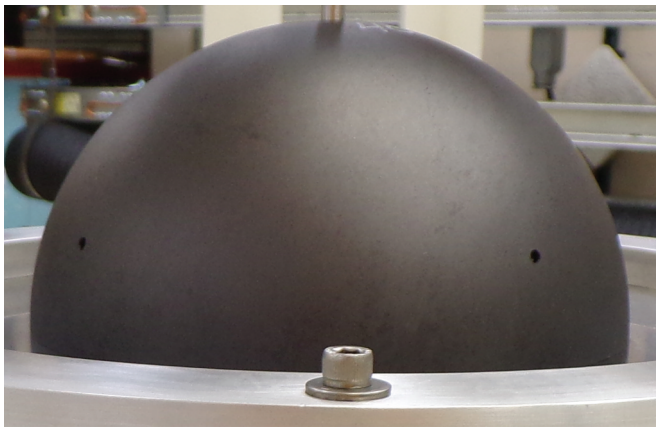


Fig. 5. A close-up picture of RF shell showing the pry holes.

II.B.3. Spacer Rings

The spacer rings were a set of flat aluminum rings that maintained a separation distance between the top and bottom stacks of a configuration when the configuration was fully assembled. Before closure, a spacer ring was placed on top of the bottom stack so that when the configuration was fully assembled, there was the desired separation between the stacks. The separation served as a reactivity control for the experiment. For the first critical configuration, the thickness of the ring was approximately 0.15 cm. The thickness of the ring for the second configuration was approximately 0.06 cm.

II.C. Miscellaneous Components

The upper stack of the RF hemishells rested on a stainless steel membrane plate. This component and the spacer ring separated the upper and lower stacks of hemishells on a fully closed configuration.

Other components that were part of both configurations were the membrane clamp plate, swivel plate, swivel mount plate, outer support bars, central support bar, upper plate standoffs, Planet top plate, Planet tilt-weigh table, etc. Figures 6 and 7 show all the components.

II.D. Experimental k_{eff}

The reactivity of the configurations was calculated using the Inhour equation defined as

$$\rho(\$) = \frac{\Lambda}{T\beta_{eff}} + \sum_{i=1}^{6} \frac{\frac{\beta_i}{\beta_{eff}}}{1+\lambda_i T} \quad , \tag{1}$$

where T = reactor period (s); β_{eff} = delayed neutron fraction; Λ = prompt neutron lifetime of the system (s); $\frac{\beta_i}{\beta_{eff}}$ = relative abundance for the i'th delayed group; λ_i = decay constant for the i'th delayed precursor group; $\frac{\beta_{eff}}{\Lambda}$ = measured Rossi-α, or the prompt neutron lifetime, at delayed critical (s^{-1}). For a fast ^{235}U system, the Ross-α is 1.1×10^6 s^{-1}. Reference [14] lists the values for each of the six groups' relative abundances and decay constants for a fast ^{235}U system.

Based on Eq. (1), the measured reactivity and associated standard deviation for each case are reported in Table V. The standard deviations reported in Table V are based on the measured reactor periods for all fully assembled configurations as well as the variation measured when unstacking and restacking both configurations.

To convert reactivity to k_{eff}, a β_{eff} value of 0.0065 was used, which is a measured value from Lady Godiva.[14] For comparison, the calculated β_{eff} value reported from simulations is 0.00652 ± 0.00002 for Configuration 8 and 0.00651 ± 0.00002 for Configuration 9. To assess any potential uncertainty in the delayed neutron parameters, the β_{eff} value was increased 15% up to 0.0074. This is meant to bound any possible delayed neutron parameter that could be used. Since the nominal value of 0.0065 is at one end of the bounds being considered as part of this uncertainty, the uncertainty in k_{eff}, u_k, is derived by dividing by $\sqrt{6}$. The results of these perturbations are reported in Sec. III.

III. EVALUATION OF UNCERTAINTIES

The sensitivity and uncertainty effects for the two configurations were evaluated using the Monte Carlo N-Particle Version 6.2.0 code[15] and ENDF/B-VIII.0 neutron cross-section libraries. The MCNP simulations were completed using 500 000 histories per generation with 3400 active generations. The statistical 1σ uncertainty associated with all Monte Carlo simulations is ±0.00001 or ±1 pcm.

For some parameters, the uncertainty in k_{eff}, $u_{k,i}$, due to the uncertainty in parameter i with reference value $x_{i,0}$ was calculated as the difference between perturbed values of k_{eff} divided by the number of standard deviations, as seen in Eq. (2):

$$u_{k,i} = \frac{u_i}{2\delta x_i}\left[k_{eff}\left(x_{i,0}+\delta x_i\right) - k_{eff}\left(x_{i,0}-\delta x_i\right)\right] \quad , \tag{2}$$

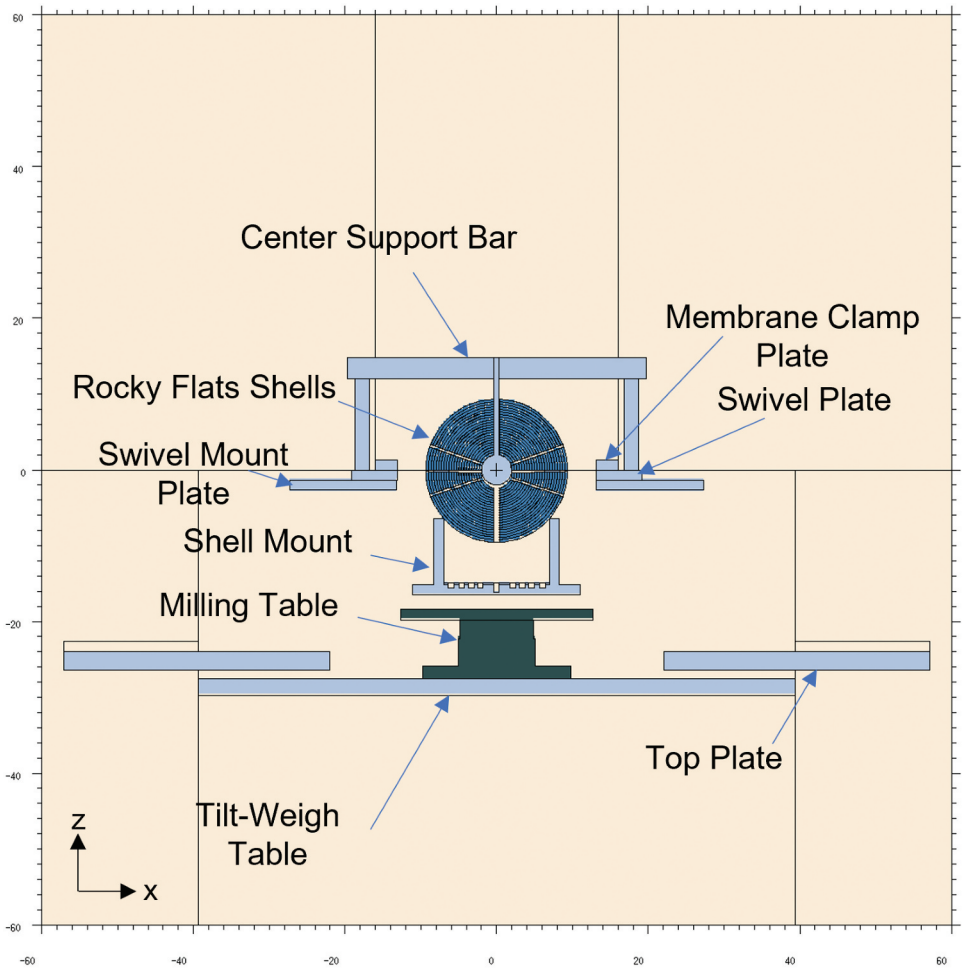


Fig. 6. An overview of the detailed model with components labeled. The scale is in centimeters.

where u_i = uncertainty in the parameter being perturbed; δx_i = value of the perturbation. The scaling factor $2\delta x_i/u_i$ is the number of standard deviations by which x_i is being perturbed. The difference in Eq. (2) is given as Δk_{eff} in the tables in Sec. IV. This Δk_{eff} is converted to an uncertainty (reported as ± a value) via the $2\delta x_i/u_i$ scaling factor for a two-sided uncertainty. For example, a Δk_{eff} of 150 pcm from perturbing a parameter by three standard deviations in each direction turns into a ±25 pcm uncertainty. However, if the perturbation in question is one-sided, then the factor of 2 is omitted.

For other parameters, such as mass and composition, it is possible to use the adjoint-based relative sensitivity coefficients to compute all the associated uncertainties with a single calculation for each configuration. This is done through the KSEN card in MCNP.[16,17] Massive computational and user time is saved in this way, as many input files with perturbations in mass or compositions for specific parts, or nuclides within those parts, do not need to be made or run. For example, normally, one would run two input files for each component to tweak the mass, and twice as many for each component as there are impurities in the composition in order to get a central-difference estimate of the uncertainty in the k_{eff} values. Running one simulation instead of many dozens, and also not having to perturb the input deck (or devote time to automating these perturbations), is what leads to these time savings. The adjoint-based methods provide a relative sensitivity coefficient $S_{k,x}$ defined by

$$S_{k,x_i} \equiv \frac{x_{i,0}}{k_{eff,0}} \frac{\delta k_{eff}}{\delta x_i}\bigg|_{x_i = x_{i,0}} . \quad (3)$$

This can then be combined with the uncertainty in the parameter (e.g., mass, composition) to get the u_k through Eq. (4)[18]:

$$u_{k,i} = \pm \frac{u_i}{x_{i,0}} k_{eff,0} S_{k,i} . \quad (4)$$

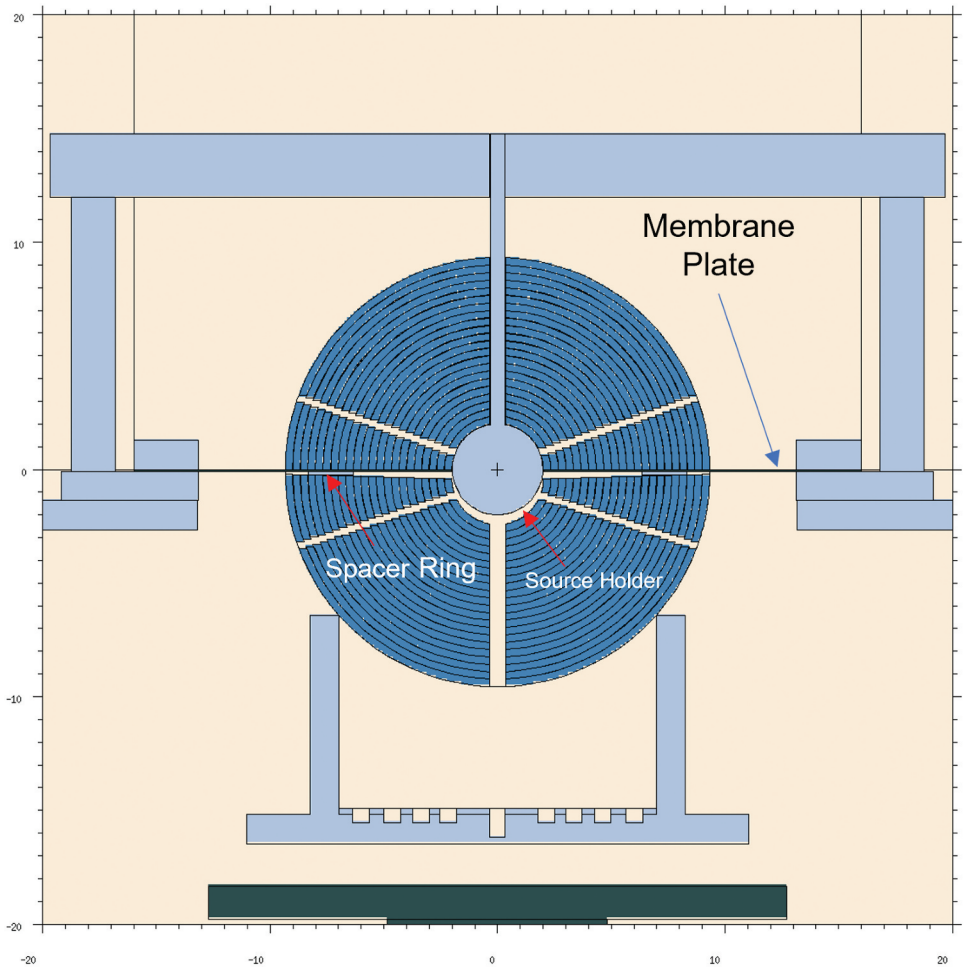


Fig. 7. A narrower view of the detailed model. The scale is in centimeters.

TABLE V

Measured Reactivity for Each of the Two Configurations

Case	Reactor Period (s)	Reactivity (¢)	k_{eff}
1	14.23 ± 0.09	32.88 ± 0.11	1.00211 ± 0.00001
2	13.82 ± 0.78	33.38 ± 0.91	1.00214 ± 0.00006

The uncertainties associated with these experiments have been divided into six categories: critical measurements, mass, dimensions of components, material composition, positioning/surroundings, and temperature. Each category is evaluated separately and later combined to yield the total experiment uncertainty. Each estimated uncertainty reported in Table VI is one standard deviation.

The first category, critical measurements, explains how the reactivity and the k_{eff} of each of the two configurations was evaluated. Section II.D discusses how the reactivity was calculated based on the measured reactor period. The uncertainties associated with the reactor period and the delayed neutron fraction used to convert reactivity to k_{eff} are also discussed in Sec. II.D.

The second category evaluated was the uncertainties associated with the uncertainty of the mass. As mentioned above, the mass uncertainty was estimated using the adjoint-based sensitivity coefficients (KSEN card in MCNP) and Eqs. (3) and (4). The mass uncertainty of every component was analyzed, including the RF hemishells, other core components, and the structural materials surrounding the assembled configurations.

The third category evaluated was the uncertainties associated with the dimensions of the core components. For these simulations, direct perturbations were performed adjusting the dimension of each component while the mass of the component was held constant.

The fourth category evaluated was the material composition (e.g., isotopics and impurities) for each component. Similar to the mass, these uncertainties were

TABLE VI

The Magnitudes of u_k for the Categories of Uncertainties

Category	Case 1	Case 2
Critical measurement	−14/+19 pcm	−15/+25 pcm
Rocky Flats mass	±6 cm	±6 pcm
Rocky Flats dimensions	±0 pcm	±84 pcm
Rocky Flats composition	±59 pcm	±59 pcm
Aluminum core components positioning	±27 pcm	±27 pcm
Temperature	±10 pcm	±10 pcm
Total	±120 pcm	−116/+118 pcm

calculated using the adjoint-based approach of Eqs. (3) and (4) and the KSEN card in MCNP.

The fifth category evaluated was the uncertainty in the positioning of each of the core components. For example, this category examined possible errors in the location of the RF hemishells, the membrane, and other components around the assembled configurations.

Finally, the sixth category evaluated was the uncertainty in the temperature of the configurations and its surroundings.

Table VI shows the resulting uncertainties u_k associated with each category for each configuration. As expected, in two of the six categories (dimensions and composition), the largest uncertainties were associated with the perturbation of dimensions and composition of the fissile material. Other uncertainties associated with the dimension and composition perturbations of other components such as the aluminum spacer, the stainless steel membrane, or the milling table were much smaller when compared to the perturbation of the dimensions and composition of the RF hemishells. The same applies to mass where the perturbation of the fissile material mass (RF hemishells) in this category is the dominant uncertainty compared to the mass perturbation of nonfissile materials that are part of the configuration.

Finally, the total uncertainty reported in Table VI (±120 pcm for Case 1 and −116/+118 pcm for Case 2) compares quite well with the total uncertainty of the Lady Godiva benchmark evaluation of ±100 pcm.[3] This indicates that despite the Lady Godiva benchmark having a potentially inadequate documentation and uncertainty analysis, in addition to being written up decades after it was performed, its final uncertainty values may not be too far off the mark. Reference [9] describes in much more detail the uncertainties associated with each category reported in Table VI. The total evaluated uncertainty in both configurations is also shown in Table IX.

IV. DETAILED MODEL AND SIMPLIFICATIONS

A goal of an ICSBEP evaluation is twofold. First, it is to develop a model of the system that is detailed enough such that any dimensions, mass, or other details of a component that affect the system k_{eff} are accurately represented such that any differences are due to the nuclear data. Second, the evaluator is to develop a second model that balances simplifications to the system with their effect on k_{eff} such that a nuclear data evaluator or criticality safety professional can use them in a streamlined manner to test their data or methods. The evaluator may judge some parts of the system, or simplifications thereof, to be negligible; however, these judgments need to be agreed upon by the TRG. Simplifications can be made to this model for the sake of convenience; however, those simplifications must be quantified to ensure that the overall bias in k_{eff} or in its sensitivities does not become too large. The drawback of a simplified model is that the simplifications can introduce bias effects that could obscure the actual sensitivity of the model to the nuclear data.

For the MUSIC benchmark, a series of increasingly simplified models was created to analyze these effects. First, a model was created that attempts to depict reality as closely as possible. The only omission from this model was some of the additional detector equipment. A reactivity measurement of the excess reactivity had been done both with and without this equipment for a configuration. From this, an experimental simplification effect of 36 pcm was obtained. The measurement eliminated the need to do additional modeling to faithfully represent all the additional equipment (detectors and their holders, cables, placement, etc.), saving a significant amount of time and effort.

Further simplifications were then made to this model, such as removing the concrete around the room and the structural components of Planet. The quantification of these effects was determined through simulation in MCNP6.2, and the results (together with the detector equipment effect) are given in Table VII. The model resulting from these simplifications comprises the "detailed model," which is the basis for uncertainty analysis and quantification. Views of this model are given in Figs. 6 and 7. In Table VII, "Case 1" refers to Configuration 8, and "Case 2" refers to Configuration 9. This is true for all future references to Cases 1 and 2.

For ease of use for nuclear data evaluators and those validating transport codes, a further simplified model was developed. The simplification effects were quantified by simulation. These include removing impurities, simplifying the geometry of components, and moving components such

TABLE VII

Simplification Effects on k_{eff} for the Detailed Model

Simplification	$\Delta k_{eff} \pm 1\sigma_{MC}$
Removal of additional detectors	−0.00036 ± 0.00025
Shell mount simplification	−0.00019 ± 0.00001
Removal of RF impurities	0.00001 ± 0.00001
Milling table simplification	−0.00001 ± 0.00001
Source holder simplification	+0.00002 ± 0.00001
Removal of flattop	+0.00002 ± 0.00001
Removal of screws, pins, and counterbores	−0.00002 ± 0.00001
Removal of Planet structural components	−0.00035 ± 0.00004
Removal of concrete and crane	−0.00063 ± 0.00001
Temperature Case 1	+0.00022
Temperature Case 2	+0.00023
Total Case 1	−0.00129 ± 0.00025
Total Case 2	−0.00128 ± 0.00025

that the creation of a model input file is easier. A full list of simplifications is given in Table VIII, and a view of this model is in Figs. 8 and 9. For both the detailed and the simplified models, the simplification effects are added or subtracted from the experiment k_{eff} to give the benchmark k_{eff} values. This is the value that the simulations are meant to match. Tables IX and X give the simplification effects and the resulting benchmark k_{eff} values for the detailed and simplified models, respectively.

It may be desired in some cases to have an even more simplified model. The most directly comparable benchmarks in the ICSBEP are the historical benchmarks of Lady Godiva (HEU-MET-FAST-001)[3] and the Oak Ridge ORSphere (HEU-MET-FAST-100),[4] both of which have simplified models that are one-dimensional spheres of HEU. To see if such a simplification was possible for MUSIC, models were made of a simple two-region sphere. The inner sphere is aluminum to represent the source holder present in the interior of the system. Surrounding this is a sphere of HEU to represent the hemishells. Obviously, there is a massive loss in fidelity (for example, the gaps between the hemishells and the stacks, the stainless steel membrane between the stacks, and the surrounding environment). Despite this, for Case 1, the simplification effect is surprisingly small, less than 100 pcm. For Case 2, this is not true. This may be because the case involves switching out one of the shells on the bottom stack with a mock version, meaning the overall assembly is less symmetric.

Table XI gives the dimensions of the one-dimensional models and their effect on the k_{eff} of the system. The two different radii given for the Case 2 model are used to represent the inner radius of the HEU as either the inner radius of the top stack or the inner radius of the bottom stack, which has one fewer shell. Figures 10, 11, and 12 then give the cross-section sensitivities of these models in comparison with the base cases.

Perhaps, it should not be surprising that the cross-section sensitivities for a bare system are so similar despite simplifications. In fact, the E_{sum} and c_k values between all the given configurations are 1, as will be expanded upon in Sec. V. Still, the resilience of this similarity to simplification is largely due to the bare nature of the system, and future evaluations of other systems will have to prove that such a simplified model is representative.

Some simplifications that could be seen as additional steps between the fully detailed and the one-dimensional models were also attempted but ultimately not incorporated into the simplified model because the effects were too large. This includes what could be seen as natural changes such as homogenizing each stack of shells,

TABLE VIII

Simplification Effects on k_{eff} for the Simplified Model

Simplification	Case 1, $\Delta k_{eff} \pm 1\sigma_{MC}$	Case 2, $\Delta k_{eff} \pm 1\sigma_{MC}$
Previous simplifications	−0.00129 ± 0.00025	−0.00128 ± 0.00025
Removal of air	−0.00017 ± 0.00001	−0.00017 ± 0.00001
Removal of aluminum impurities	−0.00019 ± 0.00001	−0.00032 ± 0.00001
Removal of milling table keyways	+0.00020 ± 0.00001	+0.00027 ± 0.00001
Removal of shell mount channels	+0.00005 ± 0.00001	+0.00011 ± 0.00001
Removal of lower RF offsets	+0.00077 ± 0.00001	+0.00091 ± 0.00001
Mock aluminum shell simplification	–	+0.00003 ± 0.00001
Rocky Flats pry hole simplifications	+0.00009 ± 0.00001	+0.00009 ± 0.00001
Total case	−0.00054 ± 0.00025	−0.00036 ± 0.00025

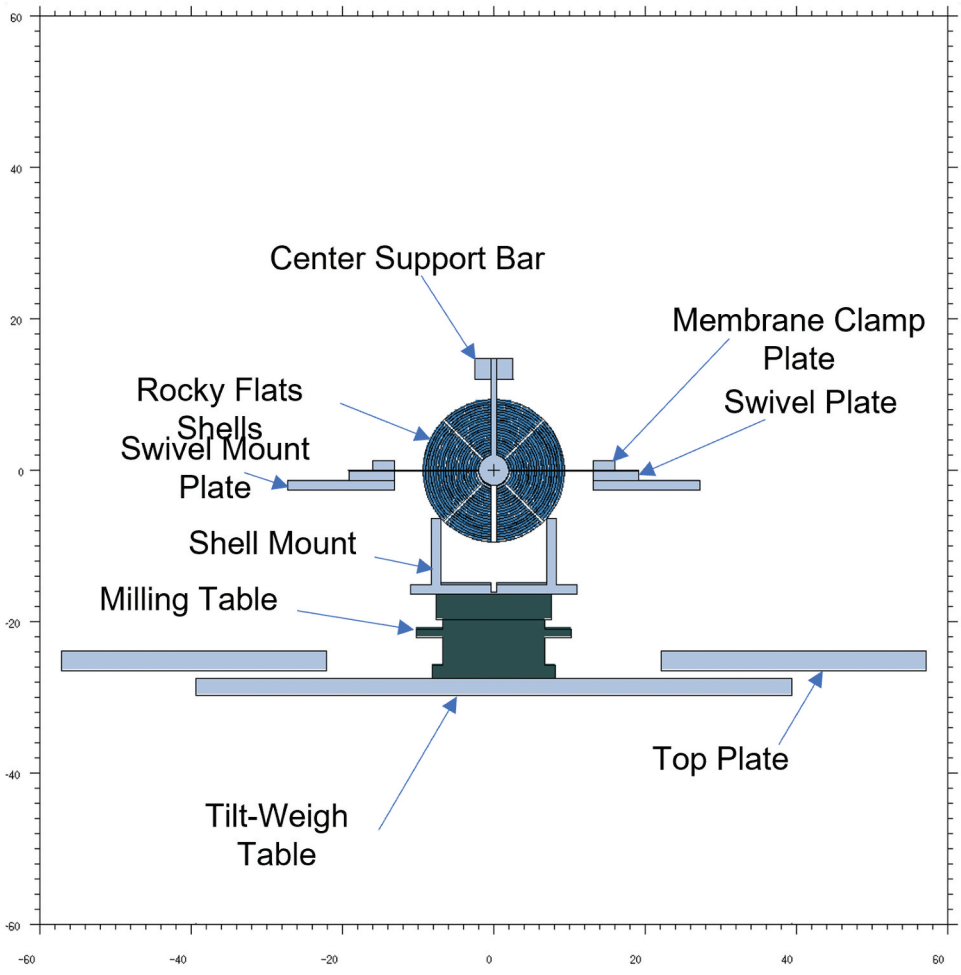


Fig. 8. An overview of the simplified model. The scale is in centimeters.

removing the pry holes in the shells, or using an average shell density. The average shell density caused k_{eff} to change by 45 to 50 pcm depending on the case while removing the pry holes and homogenizing the stacks were worth ~150 pcm each.

Simulations were performed with various cross-section libraries to assess how well they match the experiment. The results are given in Table XII for ENDF/B-VII.1, ENDF/B-VIII.0, JEFF-3.3, JENDL-5.0, and CENDL-3.2.[19–22] Figure 13 also shows this visually. All of the libraries underpredict k_{eff} to a varying extent, although all are within 200 pcm.

Another set of calculations involved swapping out individual nuclide libraries from ENDF/B-VIII.0 to other sets. In other words, for these calculations, all nuclear data come from ENDF/B-VIII.0 except for a nuclide of interest, for which the data of another library were used. The results of these calculations are given in Tables XIII and XIV. An interesting note is that while for ENDF/B-VII.1 and JEFF-3.3, ^{235}U and ^{238}U are roughly equal contributors, JENDL-5.0 is dominated by ^{235}U, and the small changes in CENDL-3.2 are dominated by ^{27}Al.

V. NUCLEAR DATA SENSITIVITY AND UNCERTAINTY

Simulations of the MUSIC configurations were performed using both ENDF/B-VII.1 and ENDF/B-VIII.0 nuclear data libraries in MCNP6.2. A hypothetical model was made of a bare sphere of 100% ^{235}U metal. The mass of the sphere was different for the two nuclear data libraries because it was based on the simulated critical masses for each nuclear data library. Table XV shows specifications for these models, and it can be seen that they are very similar to each other and have simulated k_{eff} near 1.

The sensitivity to k_{eff} was simulated using the MCNP code. Here, the sensitivity to k_{eff}, $S_{k,x}$, is defined as

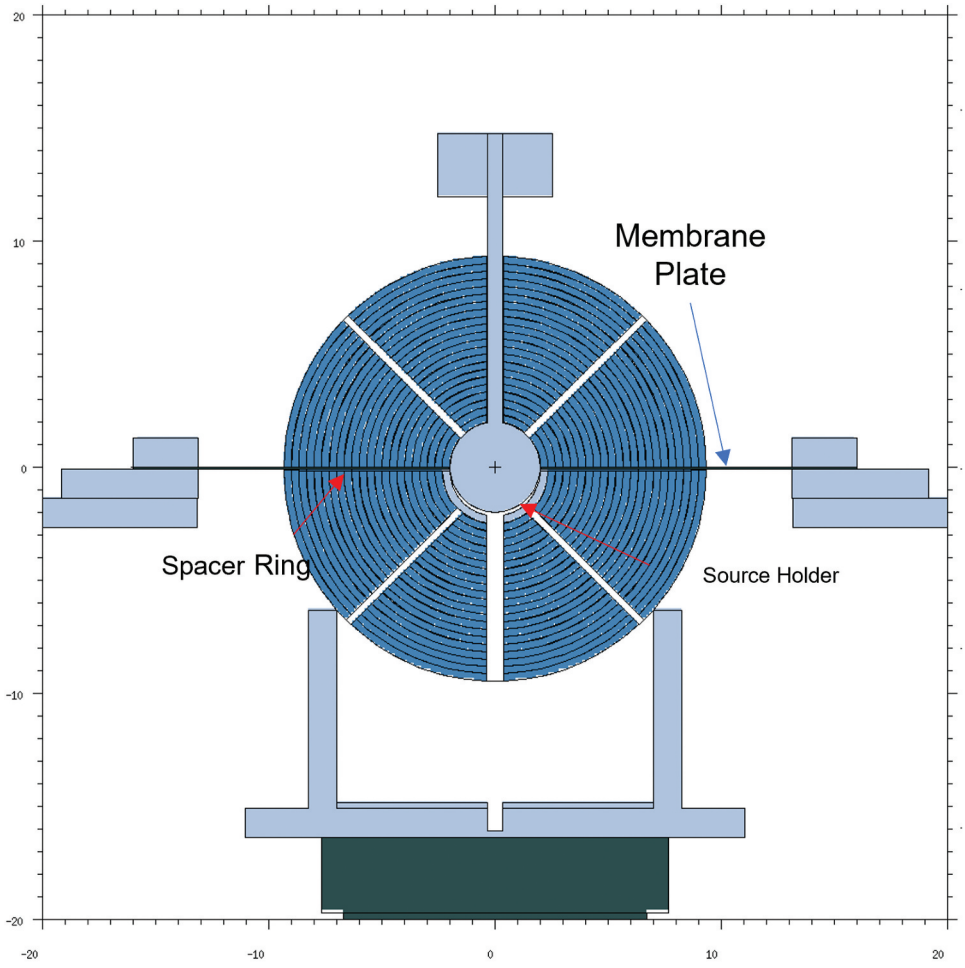


Fig. 9. A narrower view of the simplified model. The scale is in centimeters.

TABLE IX

Summary of Experimental and Benchmark k_{eff}, Uncertainty, and Bias for the Detailed Model

Case	Experiment k_{eff}	Evaluated Uncertainty	Simplification Effects	Benchmark Model k_{eff}
1	1.00211	± 0.00120	−0.00129 ± 0.00025	1.00082 ± 0.00123
2	1.00214	−0.00116/+0.00118	−0.00128 ± 0.00025	1.00086 − 0.00119/+0.00121

TABLE X

Summary of Experimental and Benchmark k_{eff}, Uncertainty, and Bias for the Simplified Model

Case	Experiment k_{eff}	Evaluated Uncertainty	Simplification Effects	Benchmark Model k_{eff}
1	1.00211	± 0.00120	−0.00054 ± 0.00025	1.00157 ± 0.00123
2	1.00214	−0.00116/+0.00118	−0.00036 ± 0.00025	1.00178 − 0.00119/+0.00121

TABLE XI

Calculated Results for the One-Dimensional Models

Case	$k_{eff} \pm 1\sigma_{MC}$	Simplification Effect	C-E
Case 1	1.00077 ± 0.00001	−0.00083	−0.00051
Case 2, 2.3377 cm	0.99435 ± 0.00001	−0.00714	−0.00065
Case 2, 2.0126 cm	0.99832 ± 0.00001	−0.00317	−0.00065

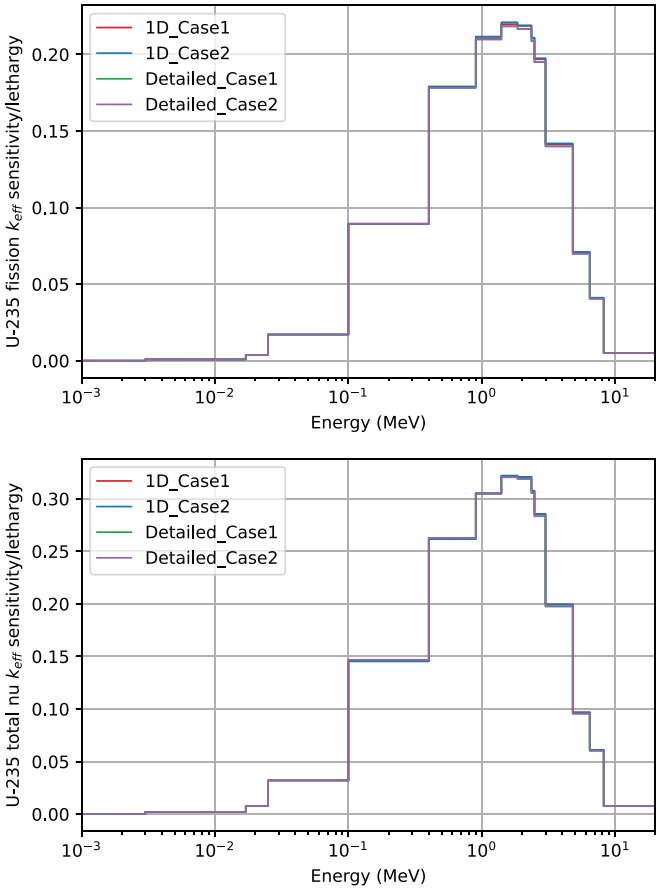


Fig. 10. The comparison of cross-section sensitivities between the detailed and one-dimensional models for fission (top) and total nubar (bottom).

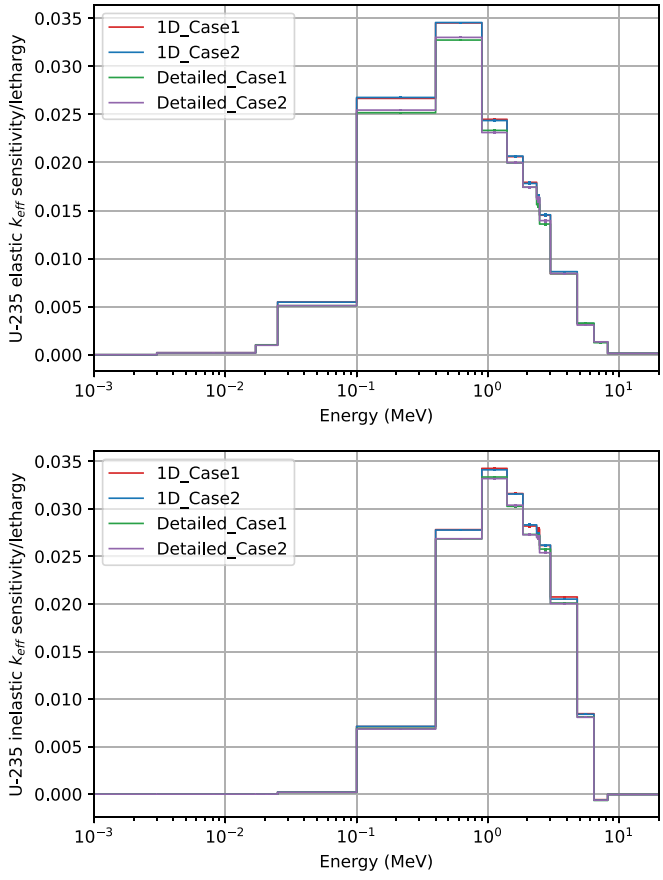


Fig. 11. The comparison of cross-section sensitivities between the detailed and one-dimensional models for elastic (top) and inelastic (bottom) scattering.

$$S_{k,x} = \frac{\partial k_{eff} / k_{eff}}{\partial x / x} \quad , \tag{5}$$

where x is a nuclear datum observable (reaction cross section, nubar, etc.) of interest. Sensitivities are shown in Figs. 14 through 20. These figures include the bare ^{235}U sphere (labeled "App U235") and four benchmark models: Lady Godiva (HEU-MET-FAST-001), ORSphere (HEU-MET-FAST-100), MUSIC Case 1 (HEU-MET-FAST-104-001), and MUSIC Case 2 (HEU-MET-FAST-104-002). HEU-MET-FAST-001 was chosen because it has been used as the primary benchmark experiment for ^{235}U nuclear data validation for 70 years. HEU-MET-FAST-100 was included as it is a more modern benchmark but also utilizes a simple HEU bare spherical design, similar to HEU-MET-FAST-001 and MUSIC.

Figures 14 through 20 have sensitivities in each top subplot along with residuals on the bottom subplot. For fission (Fig. 15) and total $\bar{\nu}$ (Fig. 14), it can be seen that the ^{235}U sphere application has slightly higher sensitivities than all of the benchmarks between 1 and 10 MeV. Out of these benchmarks, MUSIC does have slightly

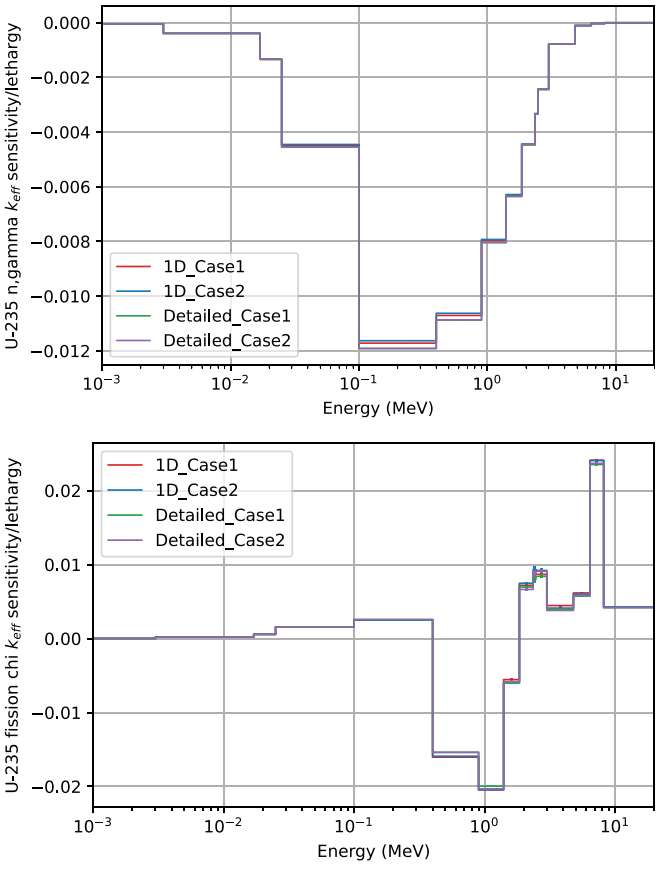


Fig. 12. The comparison of cross-section sensitivities between the detailed and one-dimensional models for (n,γ) (top) and fission χ (bottom).

worse agreement than HEU-MET-FAST-001 and HEU-MET-FAST-100. That being said, it can be seen in Figs. 14 through 20 that the MUSIC benchmark has very similar sensitivities. In general, all of the ^{235}U sensitivities are fairly similar among the benchmarks investigated.

As shown in Fig. 20, there is sensitivity to ^{27}Al elastic scattering for MUSIC. It should be noted, however, that the magnitude of the sensitivities in Fig. 20 is much smaller than those of the reactions shown in Figs. 14 through 18.

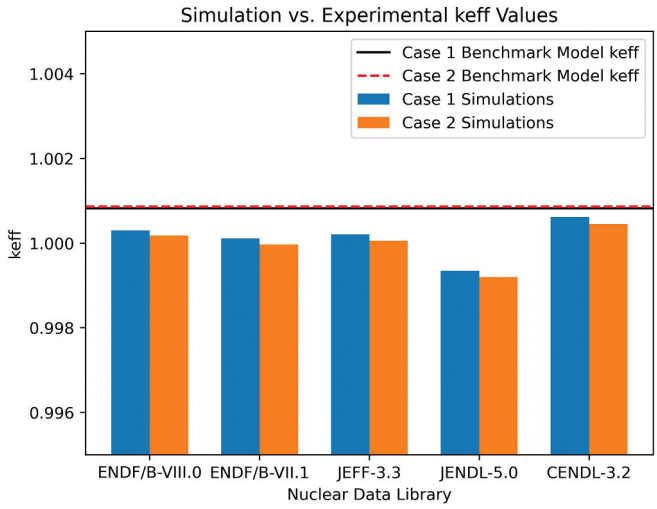


Fig. 13. The agreement between simulated and experimental results for different nuclear data libraries.

Figure 21 shows the energy-integrated sensitivities for several nuclide-reaction pairs. It can be seen that all of the benchmarks have very similar energy-integrated sensitivities to the ^{235}U bare metal sphere. In addition, it can be seen that the nuclides other than ^{235}U have very small sensitivities.

The simple similarity metric E_{sum} involves only k_{eff} sensitivities. It is defined as

$$E_{sum}(A,B) = \frac{\vec{S}_A \vec{S}_B^T}{\sqrt{\vec{S}_A \vec{S}_A^T}\sqrt{\vec{S}_B \vec{S}_B^T}} \quad , \tag{6}$$

where $\vec{S}$ = vector of all sensitivities; T = transpose of the vector; A = application model; B = benchmark model. One useful aspect about the E_{sum} metric is that it is not reliant on the covariance data. An E_{sum} value of 1.0 means that the application and benchmark are identical in regard to nuclear data sensitivities. A value of 0 means they are essentially uncorrelated, and a value of −1.0 means they are perfectly anticorrelated. Figure 22 shows

TABLE XII

Simulated Results for the Detailed Model with Different Nuclear Data Libraries

Library	Case 1, k_{eff}	C-E (pcm)	Case 2, k_{eff}	C-E (pcm)
ENDF/B-VII.1	1.00011 ± 0.00001	−71	0.99997 ± 0.00001	−89
ENDF/B-VIII.0	1.00029 ± 0.00001	−53	1.00019 ± 0.00001	−67
JEFF-3.3	1.00021 ± 0.00001	−61	1.00006 ± 0.00001	−80
JENDL-5.0	0.99935 ± 0.00001	−147	0.99920 ± 0.00001	−166
CENDL-3.2	1.00062 ± 0.00001	−20	1.00045 ± 0.00001	−41

TABLE XIII

Change in k_{eff} Values from Swapping Individual Nuclide Libraries from ENDF/B-VIII.0 for Case 1

Library	^{235}U (pcm)	^{238}U (pcm)	^{27}Al (pcm)
ENDF/B-VII.1	−34	22	1
JEFF-3.3	−29	29	−5
JENDL-5.0	−109	12	−5
CENDL-3.2	7	9	22

TABLE XIV

Change in k_{eff} Values from Swapping Individual Nuclide Libraries from ENDF/B-VIII.0 for Case 2

Library	^{235}U (pcm)	^{238}U (pcm)	^{27}Al (pcm)
ENDF/B-VII.1	−40	22	1
JEFF-3.3	−27	30	3
JENDL-5.0	−114	13	−3
CENDL-3.2	8	6	20

E_{sum} results using ENDF/B-VIII.0 nuclear data. This figure shows that all of the benchmarks included have E_{sum} values > 0.999. Given that, both MUSIC and the other benchmarks are considered nearly identical related to nuclear data sensitivities.

If nuclear data covariances are included, then it is possible to calculate nuclear data variance σ^2 associated with a specific system using Eq. (7):

$$\sigma^2 = \vec{S}\bar{C}_{xx}\vec{S}^T \quad , \tag{7}$$

where $\bar{C}_{xx}$ is the nuclear data covariance matrix. ENDF/B-VII.1 nuclear data were used for transport and the covariance library included in the Whisper[23] code; this is a combination of ENDF/B-VII.1 and BLO[b] ("Low-fidelity Covariance Project")[24] covariances. The nuclear data uncertainties (over all energies) of many ^{235}U reactions are shown in Fig. 23. Similarly, Table XVI shows these same uncertainties. Table XVII shows the same uncertainties for ENDF/B-VIII.0 cross sections and covariances, although it should be noted that these only use the diagonal (variance) terms of the covariance matrix. The tool Whisper-tk[25] was used to perform these calculations.

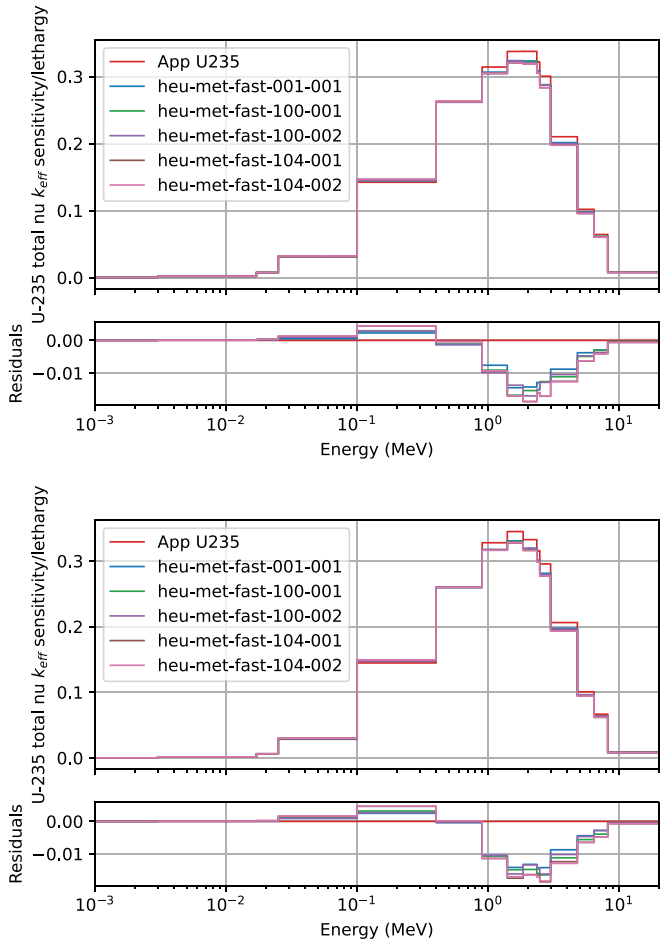


Fig. 14. Simulated sensitivities using ENDF/B-VII.1 (top) and ENDF/B-VIII.0 (bottom) nuclear data libraries for ^{235}U total $\bar{\nu}$. The ^{235}U bare metal sphere application is shown in red.

[b] Brookhaven National Laboratory (BNL)–Los Alamos National Laboratory (LANL)–Oak Ridge National Laboratory (ORNL) (BNL-LANL-ORNL): BLO.

TABLE XV

Uranium-235 Application Model Specifications

Library	^{235}U Radius (cm)	^{235}U Density (g/cm^3)	^{235}U Mass (kg)	Simulated k_{eff}
ENDF/B-VII.1	8.377254	18.92	46.6	1.00041 ± 0.00003
ENDF/B-VIII.0	8.361984	18.92	46.3	0.99928 ± 0.00003

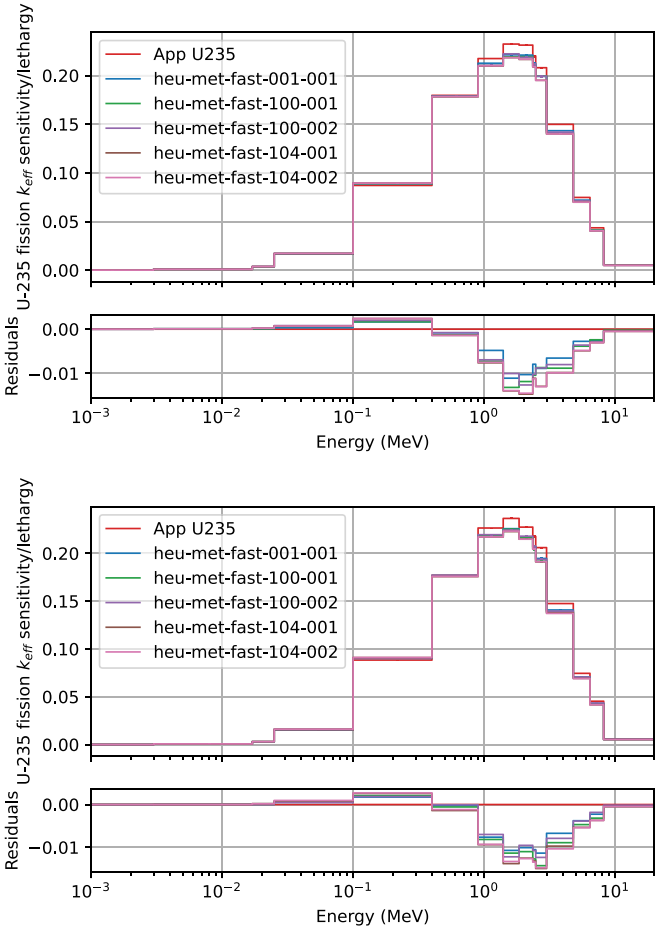


Fig. 15. Simulated sensitivities using ENDF/B-VII.1 (top) and ENDF/B-VIII.0 (bottom) nuclear data libraries for ^{235}U fission. The ^{235}U bare metal sphere application is shown in red.

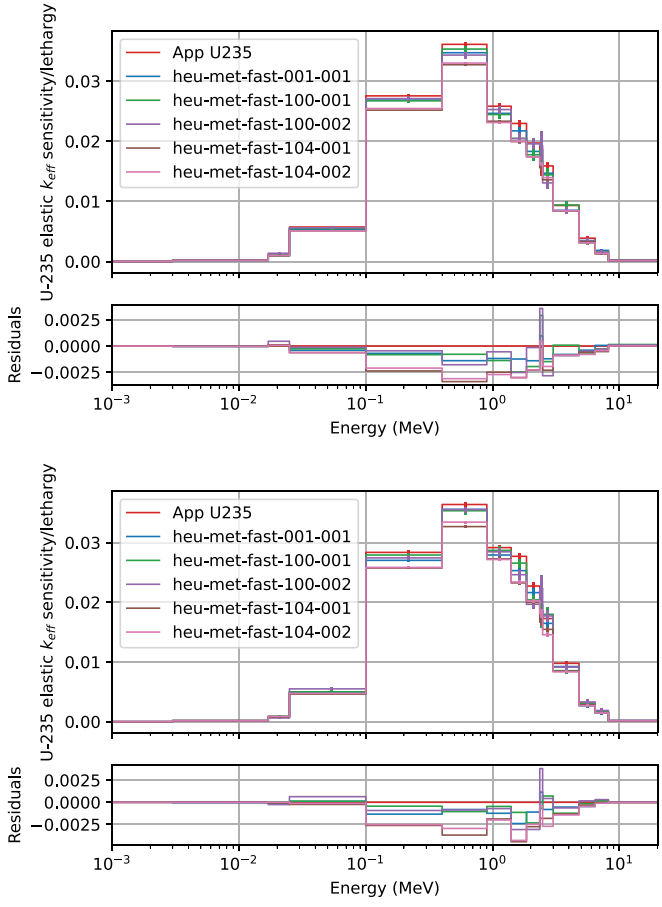


Fig. 16. Simulated sensitivities using ENDF/B-VII.1 (top) and ENDF/B-VIII.0 (bottom) nuclear data libraries for ^{235}U elastic scattering. The ^{235}U bare metal sphere application is shown in red.

It can be seen in Fig. 23 and Table XVI that ^{235}U (n,γ) is the largest contributor to nuclear data uncertainty. This agrees with previous work that looked at HEU-MET-FAST-001 as an example case.[25] It can be seen in Fig. 23 and Table XVI that all of the benchmarks investigated have very similar nuclear data uncertainties. It can also be seen that in the case of MUSIC, the nonuranium nuclides have very small nuclear data uncertainty (all $\leq$35 pcm). The total propagated nuclear data uncertainty is also very similar for all of the benchmarks (and interestingly, it is similar for both covariance data sets even though the uncertainties for specific reactions are very different).

For the ENDF/B-VIII.0 results, a few changes can be noted. First, some of the uncertainties such as ^{235}U (n,γ) and inelastic scattering go down, in addition to ^{234}U fission and ^{238}U scattering. On the contrary, uncertainties due to ^{235}U fission and elastic scattering increase.

The nuclear data covariance can also be utilized to calculate another similarity metric, c_k, which is commonly used in criticality safety validation.[23] This metric is defined as

$$c_k(A,B) = \frac{\vec{S}_A \bar{C}_{xx} \vec{S}_B^T}{\sqrt{\vec{S}_A \bar{C}_{xx} \vec{S}_A^T}\sqrt{\vec{S}_B \bar{C}_{xx} \vec{S}_B^T}} \quad , \tag{8}$$

where all terms have been defined previously. Figure 24 shows c_k results. MUSIC and all of the other benchmarks have c_k values between 0.988 and 0.990. This means that all of the benchmarks are a great fit to the ^{235}U bare sphere application and therefore are a great use case for ^{235}U nuclear data validation.

The k_{eff} calculated-to-experiment (C/E) results are shown in Fig. 25. This uses the results presented in Table X for measured values/uncertainties, Table XII for simulated values/uncertainties, and Table XVI for nuclear

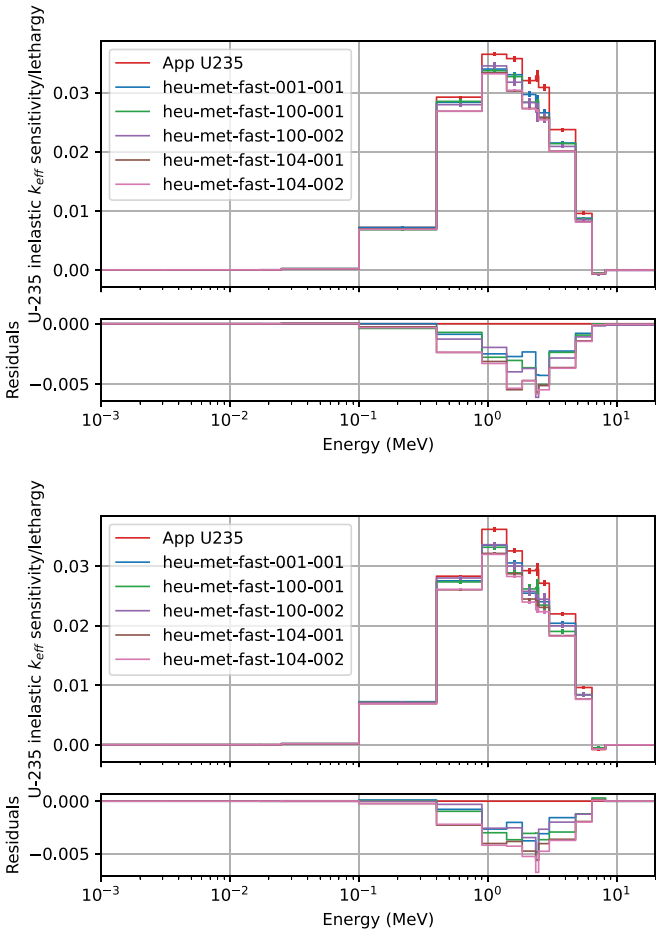


Fig. 17. Simulated sensitivities using ENDF/B-VII.1 (top) and ENDF/B-VIII.0 (bottom) nuclear data libraries for ^{235}U inelastic scattering. The ^{235}U bare metal sphere application is shown in red.

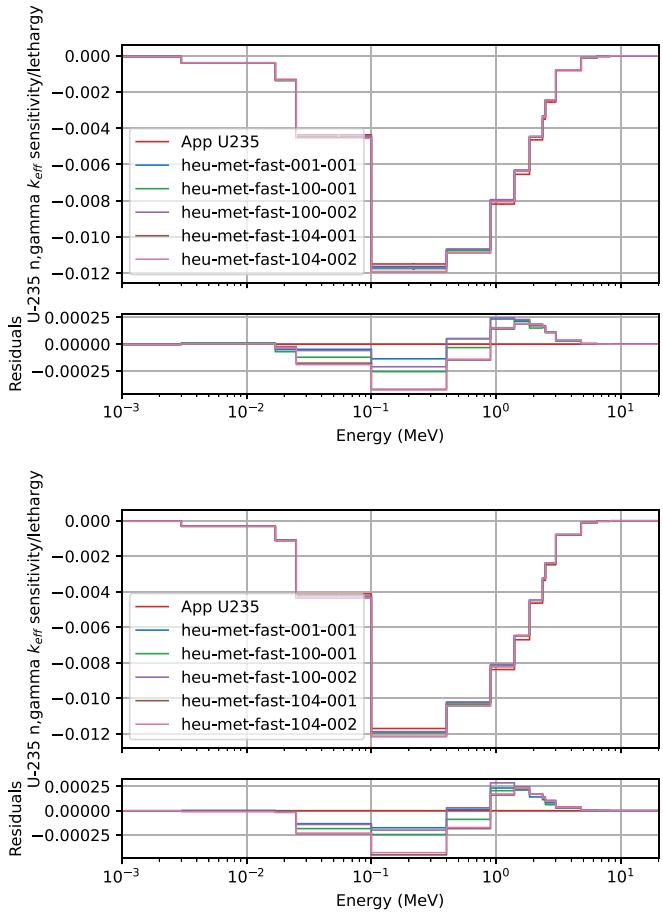


Fig. 18. Simulated sensitivities using ENDF/B-VII.1 (top) and ENDF/B-VIII.0 (bottom) nuclear data libraries for ^{235}U (n,γ). The ^{235}U bare metal sphere application is shown in red.

data uncertainties (large shaded region). This figure shows that MUSIC (in blue) simulated low (C/E < 1) and was within the one standard deviation of the experiment uncertainty for all nuclear data libraries other than JENDL-5.0. On the other hand, HEU-MET-FAST-100 simulated high (C/E > 1) and falls outside of one standard deviation of the reported experiment uncertainty. That being said, all experiments are well within one standard deviation of the much larger nuclear data uncertainties (the lighter shaded region, with results from Table XVI).

VI. SUGGESTED NUCLEAR DATA ADJUSTMENTS

For nuclear data purposes, the main use of integral experiments has historically been to validate the nuclear data evaluations. Using the experimental results to inform changes to the nuclear data has been much less common, with any comparison of performance mainly used to highlight improvements or identify potential problem evaluations. However, because the nuclear data library evaluation and validation process can be somewhat iterative for important isotopes (e.g., ^{235}U) where a benchmark like HEU-MET-FAST-001 has been used to effectively calibrate the ^{235}U evaluations over the decades, it is of general interest to observe what a more mathematically rigorous nuclear data adjustment would produce in terms of suggested nuclear data changes constrained by specific integral experiments. Some of the integral experiments studied in the present work are used to suggest potential improvements in the nuclear data while reductions in the nuclear data–induced uncertainty of an integral benchmark may also be realized.

These suggested adjustments used the generalized linear-least-squares method to minimized the χ^2 value defined by

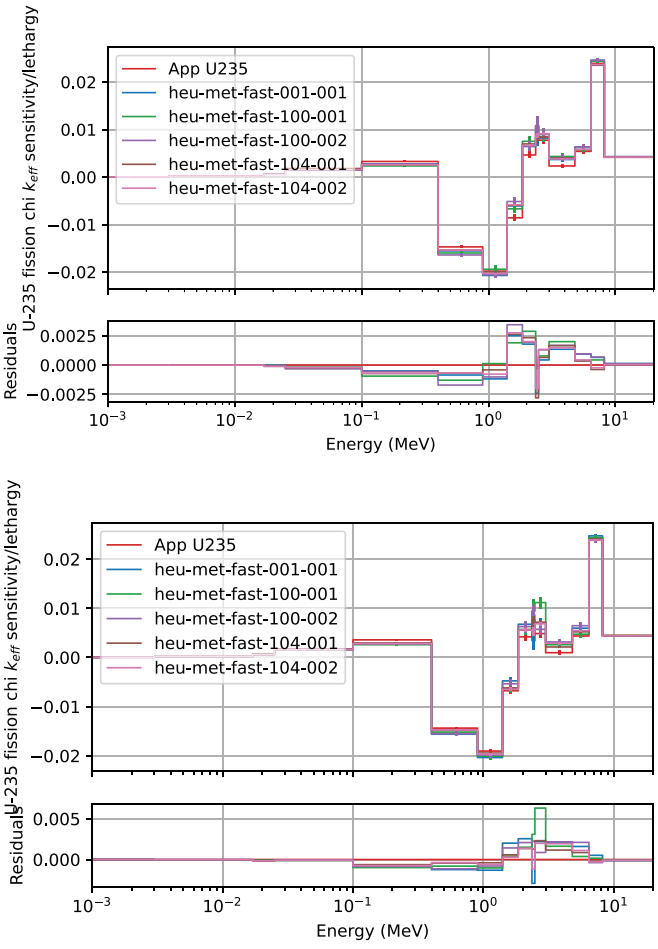


Fig. 19. Simulated sensitivities using ENDF/B-VII.1 (top) and ENDF/B-VIII.0 (bottom) nuclear data libraries for ^{235}U fission-χ. The ^{235}U bare metal sphere application is shown in red.

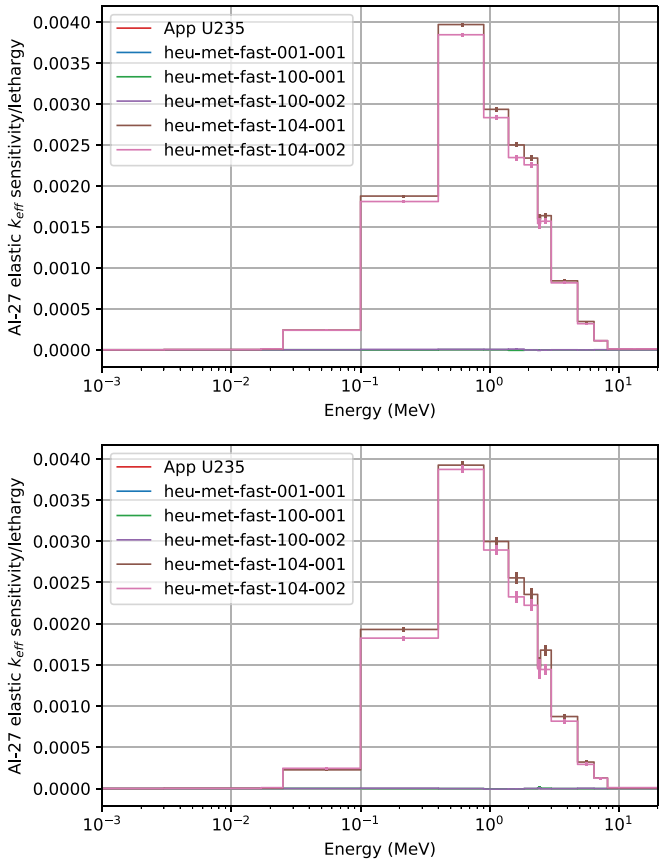


Fig. 20. Simulated sensitivities using ENDF/B-VII.1 (top) and ENDF/B-VIII.0 (bottom) nuclear data libraries for ^{27}Al elastic scattering. The ^{235}U bare metal sphere application is shown in red.

$$\chi^2 = \Delta k^T C_{mm}^{-1} \Delta k + \Delta\sigma^T C_{\sigma\sigma}^{-1} \Delta\sigma \quad , \tag{9}$$

where $\Delta\sigma$ is a vector of relative changes to the nuclear data,

$$\Delta\sigma_i = \frac{\sigma'_i - \sigma_i}{\sigma_i}, i = 1, ..., N \quad , \tag{10}$$

and Δk is a vector of relative changes in calculated k_{eff} due to changes in nuclear data.

The covariance matrices for the nuclear data and integral benchmarks are represented by $C_{\sigma\sigma}$ and C_{mm}, respectively. Posterior nuclear data changes can then be found with

$$\Delta\sigma = -C_{\sigma\sigma} S^T (S C_{\sigma\sigma} S^T + C_{mm}) d \quad , \tag{11}$$

where S = sensitivity coefficient previously discussed; d = vector of relative differences between simulated and measured k_{eff} values.[26]

For example, if ^{235}U data were adjusted to either HEU-MET-FAST-001 or HEU-MET-FAST-100, then the nuclear data uncertainty associated with that evaluation would be reduced as a result. One consequence of such adjustments, however, is that nuclear data uncertainty for a benchmark (or an application in general) could be reduced even if that benchmark was not the focus for adjustment. Figures 26 through 30 show the suggested nuclear data adjustments for ^{235}U if one adjusts to either HEU-MET-FAST-001, either of the MUSIC cases, or combinations of the HEU-MET-FAST-001 and MUSIC cases. Because the two MUSIC cases likely have a strong correlation (i.e., shared experiment uncertainties) that has yet to be estimated, the proposed adjustments do not include both MUSIC cases together as the adjustment

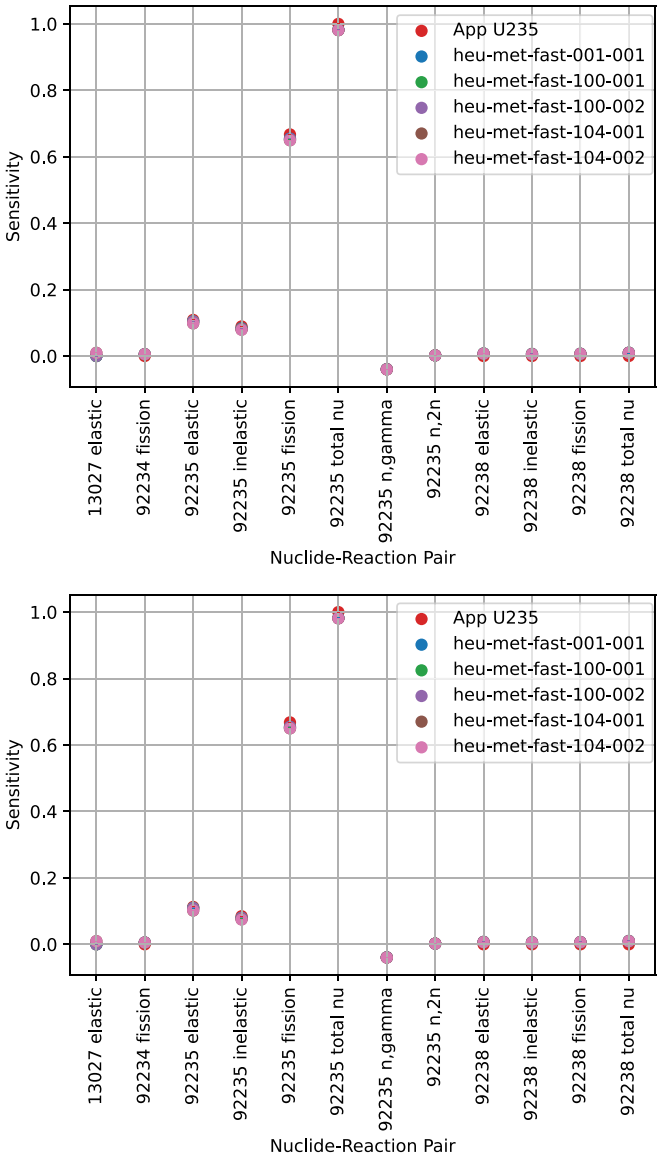


Fig. 21. Simulated sensitivities using ENDF/B-VII.1 (top) and ENDF/B-VIII.0 (bottom) nuclear data libraries summed over all energies. The ^{235}U bare metal sphere application is shown in red.

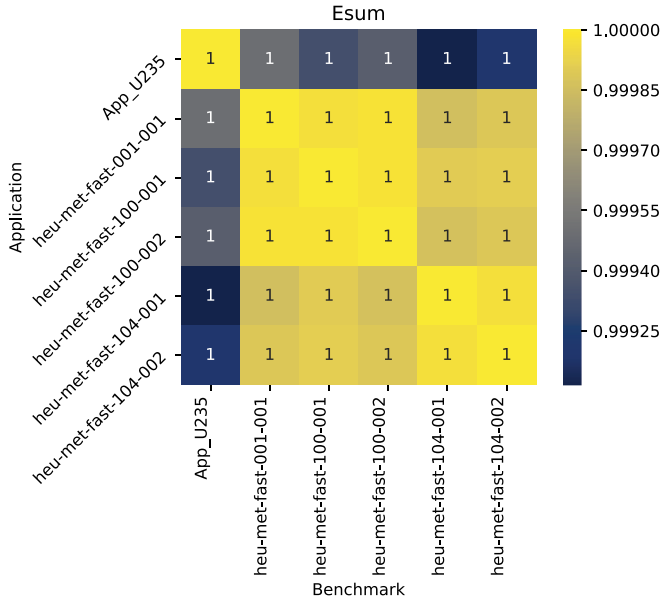


Fig. 22. The E_{sum} similarity metric between application and benchmark models using ENDF/B-VIII.0 nuclear data.

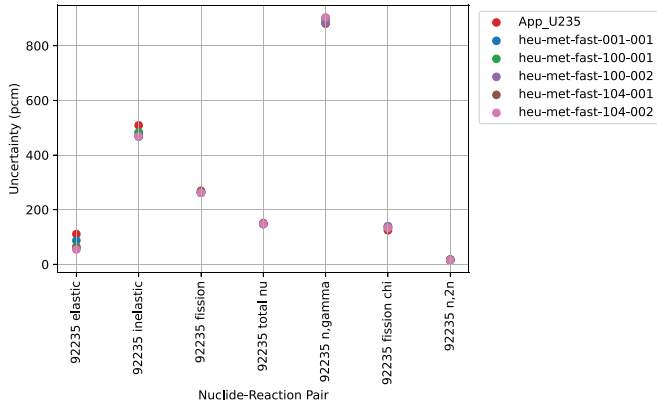


Fig. 23. Simulated nuclear data uncertainties using ENDF/B-VII.1 nuclear data libraries with ENDF/B-VII.1 and BLO nuclear data covariances. The ^{235}U bare metal sphere application is shown in red.

constraint. One thing to note is that all adjustment results are suggesting movements in the same direction, though not exactly the same magnitude. It can also be seen that the strongest suggestions for changes are to the (n,γ) interaction with elastic and inelastic also having suggested changes. In some cases, comparing the adjustments proposed in Figs. 26 and 27, where only the HEU-MET-FAST-001 and MUSIC Case 1 experiments were used individually as a constraint, to the adjustment proposed in Fig. 29, where both benchmarks were used together as the constraint, some compromises in the adjustment can be seen in certain reactions (e.g., see magnitudes of the n,γ reaction) while some changes are amplified in the adjustment to attempt to fit the nuclear data to both experiments (e.g., see χ spectrum differences). In general, the adjustments shown are reassuring that even if HEU-MET-FAST-001 does not have the most accurately quantified uncertainties, then at least a benchmark that does is in agreement with it.

Regarding the simulation biases and nuclear data associated uncertainties, Table XVIII shows the prior

TABLE XVI

Simulated Nuclear Data Uncertainties Using ENDF/B-VII.1 Nuclear Data and ENDF/B-VII.1/BLO Nuclear Data Covariances*

Nuclide	Reaction	App ^{235}U	HMF 001-001	HMF 100-001	HMF 100-002	HMF 104-001	HMF 104-002
13027	Elastic	–	–	–		4	5
13027	Inelastic	–	–	–		9	10
24052	Elastic	–	–	–	–	1	1
26056	Elastic	–	–	–		6	5
92234	Elastic	–	9	7	9	10	9
92234	Fission	–	148	143	143	145	145
92234	Inelastic	–	9	9	10	9	11
92234	*n*,γ	–	4	4	4	4	4
92235	Elastic	111	87	65	59	59	54
92235	Fission	269	264	263	264	262	262
92235	Inelastic	509	482	483	469	467	468
92235	*n*,2*n*	18	17	16	16	15	15
92235	*n*,γ	881	884	896	885	902	904
92235	Total nu	151	148	148	148	147	147
92236	Elastic	–	–	2	1	7	7
92236	Fission	–	–	3	3	28	28
92236	Inelastic	–	–	1	0	9	9
92236	*n*,γ	–	–	0	0	2	2
92238	Elastic	–	36	38	39	32	36
92238	Fission	–	3	4	4	3	3
92238	Inelastic	–	58	60	65	61	61
92238	*n*,2*n*	–	2	2	2	2	2
92238	*n*,γ	–	2	2	2	2	2
92238	Total nu	–	11	12	12	11	11
Total	–	1065	1068	1079	1065	1078	1080

*Simulated nuclear data uncertainties are in units of pcm.

TABLE XVII

Simulated Nuclear Data Uncertainties Using ENDF/B-VIII.0 Nuclear Data and ENDF/B-VIII.0 Nuclear Data Covariance Diagonal Terms*

Nuclide	Reaction	App ^{235}U	HMF 001-001	HMF 100-001	HMF 100-002	HMF 104-001	HMF 104-002
13027	Elastic	–	–	–		22	21
13027	Inelastic	–	–	–		6	7
24052	Elastic	–	–	–	–	1	1
26056	Elastic	–	–	–		2	2
92234	Elastic	–	3	3	3	2	3
92234	Fission	–	56	54	54	55	55
92234	Inelastic	–	6	6	6	7	5
92234	*n*,γ	–	5	5	5	5	5
92235	Elastic	149	142	142	141	136	136
92235	Fission	303	300	299	264	298	298
92235	Inelastic	147	138	138	469	134	131
92235	*n*,2*n*	26	23	23	16	22	21
92235	*n*,γ	164	164	166	165	167	167
92235	Total nu	159	158	158	158	158	158

(Continued)

TABLE XVII (Continued)

Nuclide	Reaction	App ^{235}U	HMF 001-001	HMF 100-001	HMF 100-002	HMF 104-001	HMF 104-002
92236	Elastic	–	–	0	0	1	1
92236	Fission	–	–	1	1	9	9
92236	Inelastic	–	–	1	1	4	4
92236	*n*,γ	–	–	0	0	1	1
92238	Elastic	–	10	11	12	11	11
92238	Fission	–	5	6	6	6	6
92238	Inelastic	–	15	16	16	15	13
92238	*n*,γ	–	1	1	1	1	1
92238	Total nu	–	5	5	5	5	5
Total	–	434	430	430	431	428	427

*Simulated nuclear data uncertainties are in units of pcm.

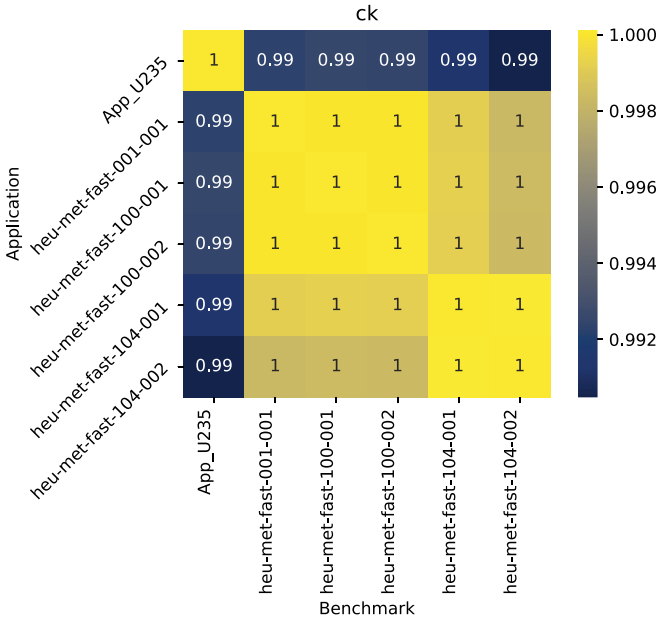


Fig. 24. The c_k similarity metric between application and benchmark models using ENDF/B-VII.1 nuclear data and ENDF/B-VII.1/BLO nuclear data covariances.

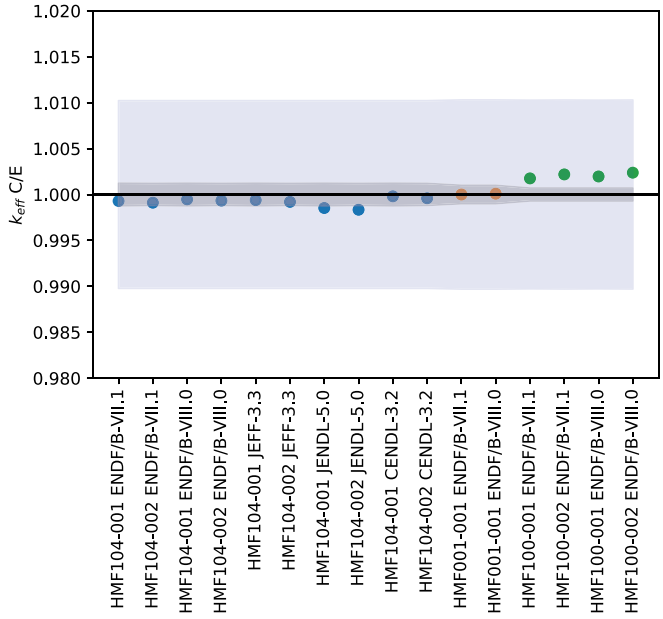


Fig. 25. The k_{eff} C/E for MUSIC (blue), HEU-MET-FAST-001 (orange), and HEU-MET-FAST-100 (green) using multiple nuclear data libraries. The experiment uncertainties are shown in the smaller (and darker) shaded region. The lighter blue shaded region is of the nuclear data uncertainties.

and posterior values for each of the adjustments shown in Figs. 26 through 30. The change in k_{eff} values can be inferred from the differences between the third and fifth columns. In general, it is also the case that the calculated k_{eff} values increased between around 59 and 253 pcm. The highest changes are due to adjusting to Case 2 of MUSIC. When adjusting to a single benchmark, the simulation biases for the independent benchmarks either show little improvement (see the HEU-MET-FAST-001 adjustment constraint row in Table XVIII) or a larger disagreement with the benchmark values (see the MUSIC Case 1 and MUSIC Case 2 adjustment constraint rows in Table XVIII). When the HEU-MET-FAST-001 and either MUSIC cases (see last two adjustment constraint rows in Table XVIII) are both used in the adjustment, nearly all simulation biases are the same or improved, and

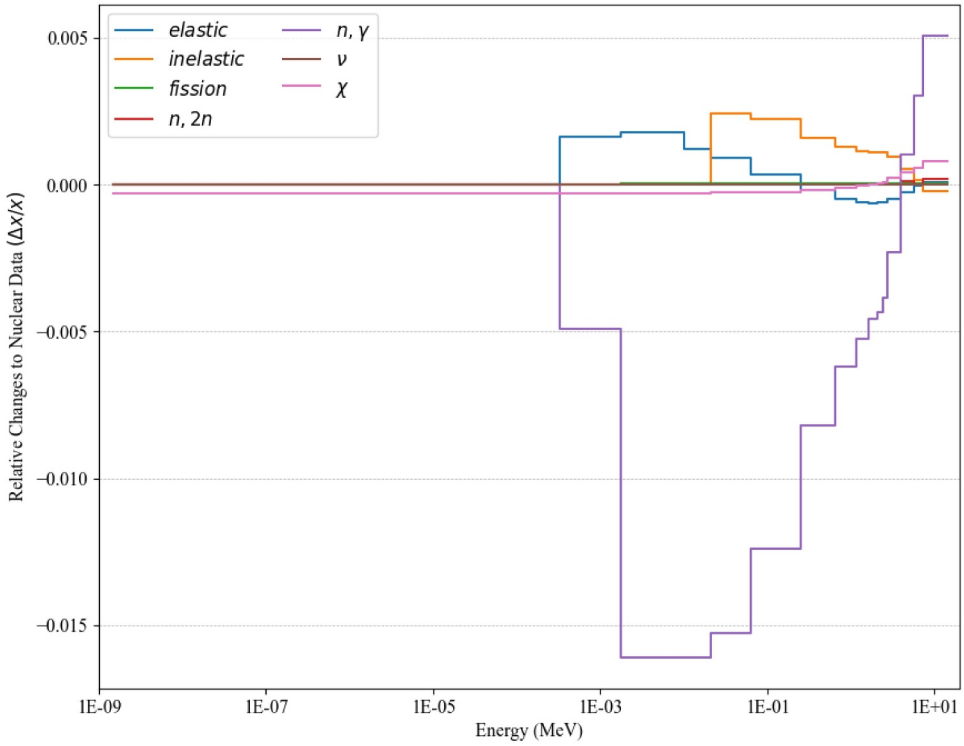


Fig. 26. Suggested ENDF/B-VII.1 nuclear data adjustments when constrained by HEU-MET-FAST-001.

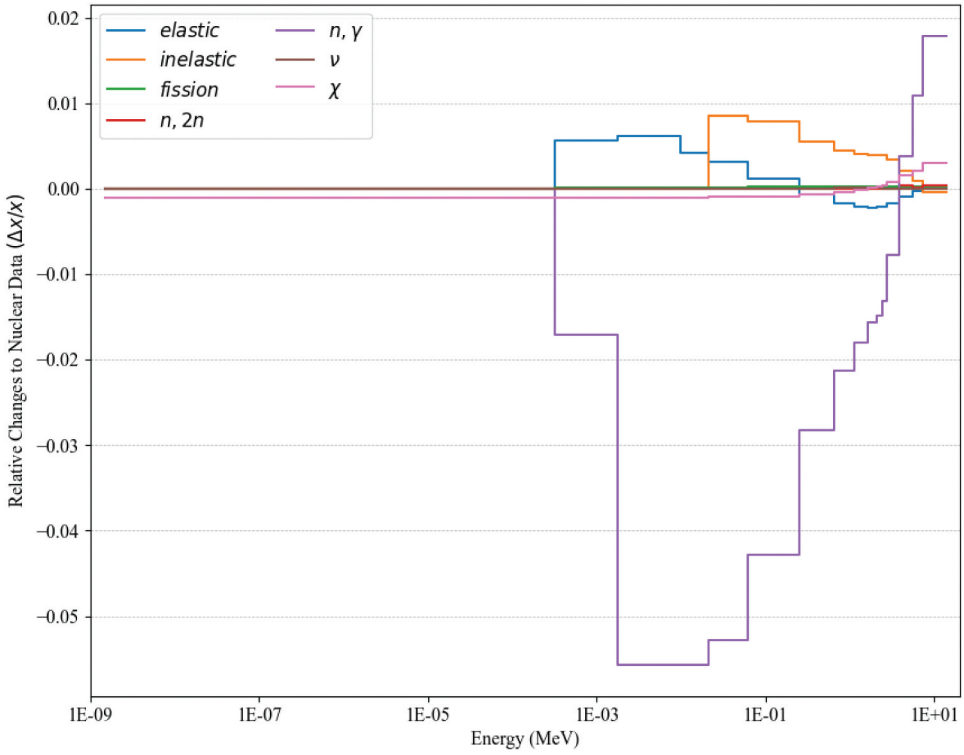


Fig. 27. Suggested ENDF/B-VII.1 nuclear data adjustments when constrained by MUSIC Case 1.

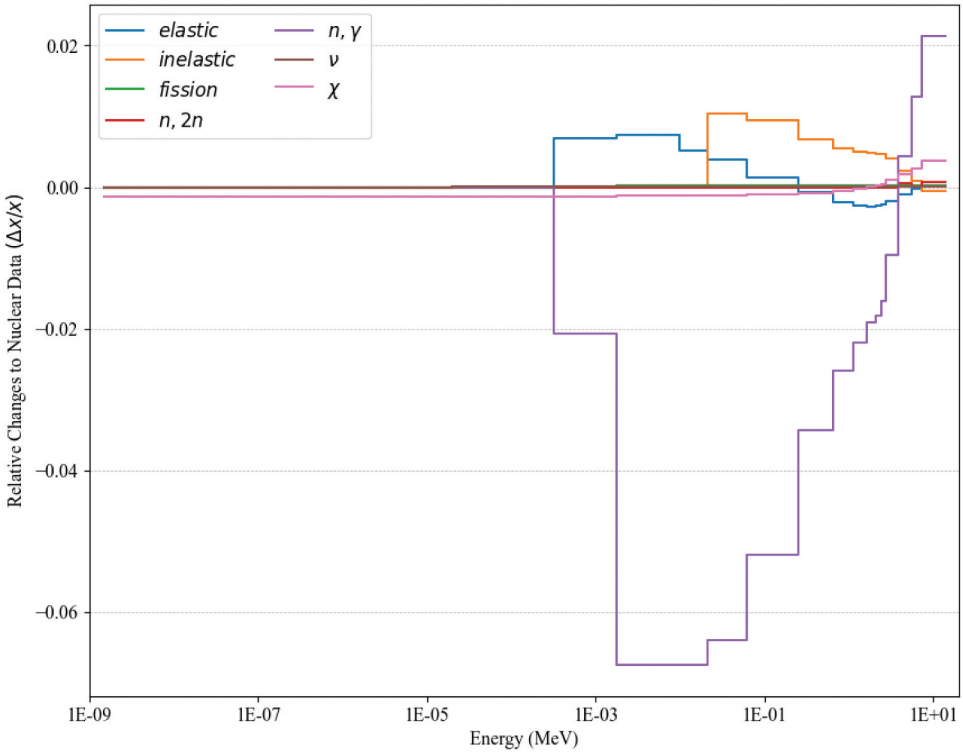


Fig. 28. Suggested ENDF/B-VII.1 nuclear data adjustments when constrained by MUSIC Case 2.

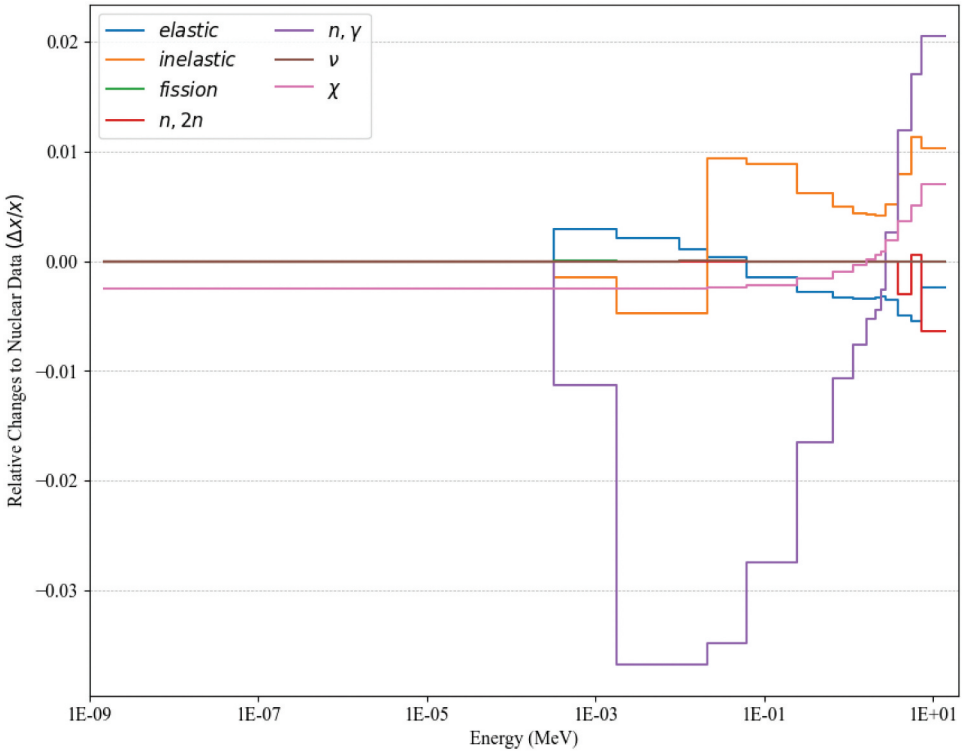


Fig. 29. Suggested ENDF/B-VII.1 nuclear data adjustments when constrained by HEU-MET-FAST-001 and MUSIC Case 1.

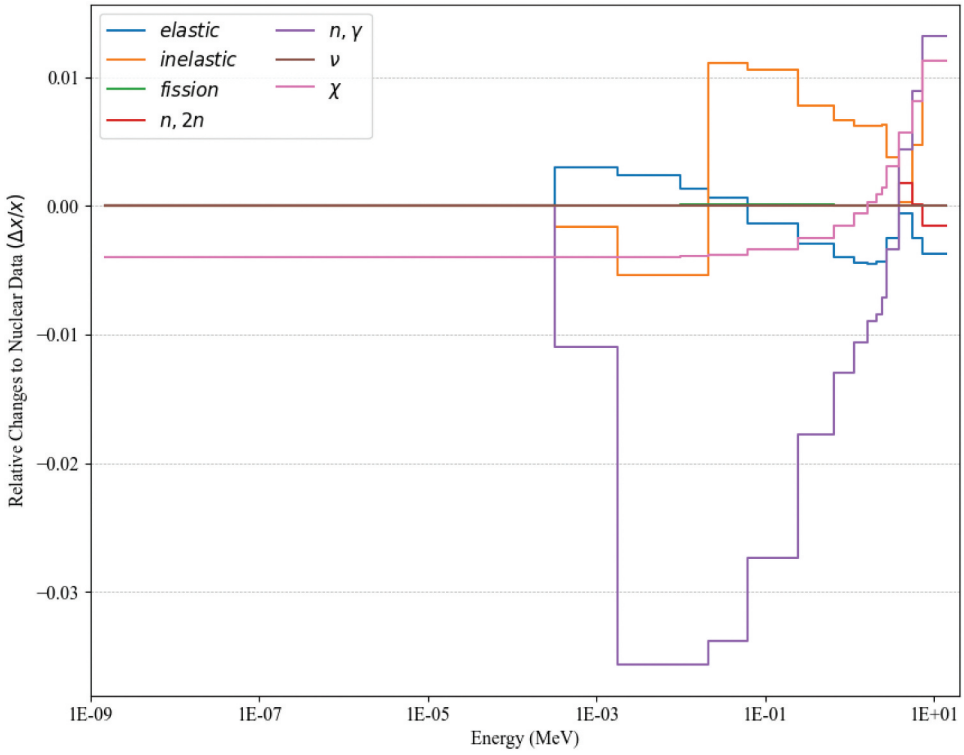


Fig. 30. Suggested ENDF/B-VII.1 nuclear data adjustments when constrained by HEU-MET-FAST-001 and MUSIC Case 2.

TABLE XVIII

Prior and Posterior Simulation Bias and Relative Nuclear Data Uncertainties When Adjusting to Various Benchmarks*

Adjustment Constraint	Benchmarks	Prior Bias (C-E in pcm)	Prior Nuclear Data Relative Uncertainties	Posterior Bias (C-E in pcm)	Posterior Nuclear Data Relative Uncertainties
HEU-MET-FAST-001	*HEU-MET-FAST-001*	*−60*	*0.01050*	*−1*	*0.00100*
	MUSIC Case 1	−211	0.01066	−151	0.00104
	MUSIC Case 2	−254	0.01057	−194	0.00102
MUSIC Case 1	HEU-MET-FAST-001	−60	0.01050	145	0.00120
	MUSIC Case 1	*−211*	*0.01066*	*−3*	*0.00119*
	MUSIC Case 2	−254	0.01057	−48	0.00119
MUSIC Case 2	HEU-MET-FAST-001	−60	0.01050	189	0.00117
	MUSIC Case 1	−211	0.01066	42	0.00118
	MUSIC Case 2	*−254*	*0.01057*	*−3*	*0.00116*
HEU-MET-FAST-001 and MUSIC Case 1	*HEU-MET-FAST-001*	*−60*	*0.01050*	*60*	*0.00077*
	MUSIC Case 1	*−211*	*0.01066*	*−86*	*0.00079*
	MUSIC Case 2	−254	0.01057	−131	0.00078
HEU-MET-FAST-001 and MUSIC Case 2	*HEU-MET-FAST-001*	*−60*	*0.01050*	*80*	*0.00076*
	MUSIC Case 1	−211	0.01066	−66	0.00079
	MUSIC Case 2	*−254*	*0.01057*	*−111*	*0.00077*

*Values in italics indicate benchmark used in adjustment.

the posterior relative uncertainty due to nuclear data is the smallest, with both values being of a comparable magnitude to the evaluated experiment uncertainties given in Tables IX and X.

VII. CONCLUSION

The MUSIC benchmark is a criticality benchmark of a simple system, performed, measured, documented, and

evaluated to the high modern standards required of the ICSBEP. The evaluation following closely in time to the experiment execution is extremely helpful in ensuring that all the necessary data and details of the experiment campaign are captured. Thorough quantification of the benchmark and simulation uncertainties, along with the impact of model simplifications, gives confidence that the results are accurate and well characterized. Major libraries are currently in good agreement (less than 200 pcm) with the experimental results, with most of the difference between libraries unsurprisingly due to ^{235}U and ^{238}U cross sections, although the major difference between ENDF/B-VIII.0 and CENDL-3.2 lies in ^{27}Al data. Many major libraries also produce simulated results within one standard deviation of the experimental results.

It was shown that the MUSIC configurations have very similar nuclear data sensitivities to HEU-MET-FAST-001 and HEU-MET-FAST-100. In addition, the nuclear data sensitivities are very similar to a fictional application model that is a bare sphere of ^{235}U metal. Both the E_{sum} and c_k similarity metrics were all ≥ 0.99. In addition, the nuclear data uncertainties for MUSIC were shown to be very small for nuclides other than ^{235}U. Suggested nuclear data adjustments were given in which the MUSIC configurations and/or HEU-MET-FAST-001 were used as the basis for adjustment. The adjusted data with singular experiments (of MUSIC or HEU-MET-FAST-001) were very similar. In addition, the posterior reductions in total nuclear data uncertainties are also very similar when adjusting to single experiments (MUSIC or HEU-MET-FAST-001). For all of these reasons, MUSIC is an excellent benchmark to be used for ^{235}U nuclear data validation and also helps ensure that the analyses performed for other bare sphere benchmarks produce uncertainties that are believable.

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